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MultiBanFakeDetect: Integrating advanced fusion techniques for multimodal detection of Bangla fake news in under-resourced contexts

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ABSTRACT

The rise of false news in recent years poses significant risks to society. As misinformation spreads rapidly, automated detection systems are essential to mitigate its impact. However, most existing methods rely solely on textual analysis, limiting their effectiveness. The challenge is further compounded by the lack of a large-scale, multimodal dataset for Bangla fake news detection, as existing datasets are either small or unimodal. To address this, we introduce **MultiBanFakeDetect**, a novel multimodal dataset integrating both textual and visual information. This dataset comprises manually curated real and fake news samples from various online sources. Additionally, we propose **MultiFusionFake**, a hybrid multimodal fake news detection framework that fuses text and image modalities using an Early Fusion approach while also comparing Late and Intermediate fusion techniques. Our experiments show that MultiFusionFake, combining DenseNet-169 and mBERT, achieves 79.69% accuracy, outperforming the text-only mBERT model's 73.13%, reflecting a 6.56 percentage point improvement. These results underscore the advantages of multimodal over unimodal methods. To the best of our knowledge, this is the first study on multimodal fake news detection in the under-resourced Bangla context, offering a promising approach to combating online misinformation.

1. Introduction

Social media have become an indispensable platform for people to share and obtain information, playing a crucial role in daily life by facilitating social interactions, boosting productivity, and improving information sharing. However, its easy access and manipulation have contribute to the proliferation of fake news. The dissemination of false information through social media not only impacts public opinion but also poses serious threats to the economy, politics, public health, and society at large. Consequently, the detection of fake news has emerged as a critical research issue, aiming to mitigate the detrimental effects of misinformation and disinformation on various aspects of modern society (Liu, Qian, Xu, Ren, & Cao, 2023; Segura-Bedmar & Alonso-Bartolome, 2022).

Massive amounts of digital data are generated every day, which makes manual fact-checking impractical for fake news identification. To enable a more rapid response, it is necessary to employ strategies that enable automatic identification of misleading information. Fake news encompasses deliberately deceptive or misleading information that is designed to misinform readers. Its production and propagation may be motivated by a number of evil intentions, such as damaging people's reputations, favoritism affecting actual events, and eroding public confidence in certain social media industries. Intentional or not, disinformation can spread uncontrollably across platforms in the form of satire, biased reporting, propaganda, or clickbait. Its main goal is usually to cause conflict or keep hoaxes alive in society. By taking advantage of the quick distribution of social media platforms, misleading information is frequently intentionally disseminated to create confusion and fear. Although many tools are available for fact-checking, it can be difficult to quickly confirm legitimacy because of the vast amount of news on many platforms (Cui & Li, 2022).

The spread of fake news and misinformation has had a significant influence on social media, journalism, and news reporting in our increasingly digitized world. How people consume online information

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has also changed. A number of serious social issues have been exacerbated by false narratives spread through these channels, such as the exaggeration of irrational worries on important occasions, such as medical crises such as the Ebola outbreak. The extensive usage of well-known social media platforms and other online sources with insufficient fact-checking tools makes it easier for people to transmit inaccurate information widely and quickly, exacerbating the spread of misinformation (Singhal, Shah, Chakraborty, Kumaraguru, & Satoh, 2019). Network immunization has become a crucial approach in mitigating the spread of fake news across online social networks (OSNs). Recent research has explored both structural and dynamic models to understand and counteract misinformation. Bhavtosh Rath and Srivastava (2020) proposed the Community Health Assessment model, which introduces the roles of neighbor, boundary, and core nodes within community structures to quantify vulnerability to fake news exposure. Their model, evaluated on Twitter datasets, showed superior performance in identifying high-risk nodes in fake news diffusion. Complementarily, Shrivastava et al. (2020) developed a differential equation-based model to simulate fake news propagation and introduced mechanisms like user verification and spreader blocking. Their analysis focused on the basic reproduction number, establishing thresholds for stability in rumor dynamics. These models collectively highlight the significance of both community structure and dynamic intervention strategies in strengthening social networks against misinformation, especially during critical events like the COVID-19 pandemic.

Fake news continues to pose a serious danger to society despite attempts to counteract it. It is not too difficult to track modified images, but it is always difficult to handle images that are utilized out of context, especially when they are paired with written descriptions that muddle categorization attempts. During situations like the COVID-19 epidemic, when approximately 65% of disinformation is disseminated by images and videos, the presence of multimedia-based misinformation is especially noticeable. False claims, for example, that cocaine effectively killed the coronavirus, became popular, and social media platforms such as Facebook, X (formerly known as Twitter), and WhatsApp played a major role in spreading these claims. This trend emphasizes how multimedia content may be strategically used to increase the exposure and popularity of misleading information, underscoring the need to address this issue in the current digital era (Hangloo & Arora, 2022).

The objective of fake news detection is to automatically ascertain whether the claims made in news articles are real. In the current media environment, where visual content is heavily weighted, adding movies or images to text creates an additional level of complexity. Fake news gains popularity and spreads more easily because of these audiovisual components, which are frequently more enticing and misleading than textual news. Studies have shown that posts with images get forwarded eleven times more often on average than those without photographs. Multimodal fake news frequently has skewed or deceptive graphics and combines both text and images.

In recent years, substantial research has been dedicated to the automatic identification of fake news. One study utilized a corpus of 57,000 Bangla news articles to tackle the challenge of detecting fake news in Bangla (Hossain et al., 2021). This research achieved high accuracies using Bi-LSTM models with GloVe and FastText embeddings, demonstrating superior performance compared to GRU. However, the approach's reliance on specific embedding techniques and a single dataset may limit its broader applicability. Another study emphasized the pervasive impact of fake news in online communities and employed SVM and MNB classifiers to identify Bangla fake news from social media (Hussain, Hasan, Rahman, Protim, & Hasan, 2020a). Despite achieving high accuracy rates, this work's focus on structured data from specific platforms may restrict its effectiveness in handling diverse sources and real-time content streams. Furthermore, the reliance on unimodal textual features through techniques like TF-IDF and CountVectorizer might overlook the contextual richness provided by visual and non-textual elements in multimedia fake news scenarios. Additionally, another comprehensive study on detecting Bangla fake news highlighted the importance of dataset balance and evaluated various deep learning models (Roy et al., 2024). While achieving remarkable accuracy with Bidirectional GRU models, this paper's reliance on a single dataset and specific model architectures may limit generalizability across different linguistic variations and evolving fake news tactics. Moreover, the unimodal focus on textual information might not fully capture the multimodal aspects crucial for detecting sophisticated fake news in multimedia environments. Another study addressed dataset quality issues by curating a refined corpus of Bangla fake news and actual news, employing diverse deep learning and machine learning algorithms (Rasel, Zihad, Sultana, & Hoque, 2022). Their reliance on specific transformer models and machine learning techniques could potentially overlook emerging fake news patterns and new media sources. Moreover, the unimodal approach primarily centered on textual analysis may not effectively leverage the complementary benefits of visual data in identifying manipulative and deceptive content prevalent in multimedia fake news. Additionally, another research explored the classification of fake news using multimodal methods combining text and image data (Segura-Bedmar & Alonso-Bartolome, 2022). While achieving high accuracy with their multimodal CNN architecture, this study's focus on curated datasets like Fakeddit may not fully capture the variability and complexities of multimedia fake news across diverse online platforms. This underscores a limitation of traditional unimodal approaches, which rely solely on textual features and may struggle to detect fake news that heavily relies on manipulative visual content and contextually rich narratives. In contrast to these unimodal approaches, another study proposed a multi-modal and multi-task learning approach integrating textual, visual, and external evidence for fake news detection (Hossain et al., 2021). Despite showing promising results in enhancing detection accuracy, this work's complexity and computational requirements may pose challenges in real-time implementation and scalability across different online platforms. Unimodal methods, which focus predominantly on textual information, may lack the capacity to effectively integrate and interpret visual and contextual signals crucial for identifying sophisticated fake news techniques prevalent in multimedia-rich environments. Another investigation explores the use of semi-supervised Generative Adversarial Networks (GANs) to fine-tune pretrained language models for classifying Bengali fake reviews (Shawon, Shahariar, Shah, Alam, & Mahbub, 2023). Detecting fake news and misinformation on social media has become increasingly important due to its serious societal impacts, such as public polarization and manipulation. Recent research emphasizes the need for context-aware approaches that analyze not only the textual content of news but also the social context in which it is shared, including user behavior, reactions, and network propagation. DANES (Truică, Apostol, & Karras, 2024) is a deep neural network ensemble architecture that integrates a Text Branch and a Social Branch to capture both content and social features, demonstrating improved accuracy on multiple real-world datasets. Similarly, GETAE (Truică, Apostol, Marogel, & Paschke, 2025) enhances fake news detection by combining a Text Branch with a Propagation Branch that models how information spreads through user networks, resulting in a novel Propagation-Enhanced Content Embedding. Transformer-based models are also leveraged in MisRoBÆRTa (Truică & Apostol, 2022), which combines BART and RoBERTa to perform multi-class misinformation detection on a large, manually verified dataset, outperforming individual transformer models. Another approach focuses on document embeddings (Truică & Apostol, 2023), showing that accurate representation of textual content can outperform more complex deep learning architectures in binary and multi-label classification tasks.

In comparison, multimodal techniques have received less attention than unimodal approaches, which mostly focus on textual information. These multimodal methods, which include textual and visual elements, have shown promising results in the identification of fake news compared to unimodal approaches. However, many of these studies have traditionally approached the challenge of identifying fake news as a binary classification task, classifying it as real or fake.

Fake images are particularly attention-grabbing because they are usually created to be visually stunning and provoke strong emotional reactions. However, the fact that they are deceptive emphasizes how crucial it is to comprehend the psychological triggers that are present in the images. Because these triggers go beyond typical object-level properties, traditional image datasets are insufficient for accurately recognizing fake images. Gathering large labeled datasets with both authentic and fraudulent images is a task that is made more difficult by the fact that human labeling and verification methods take a long time and frequently cannot keep up with the speed at which new data is being uploaded to the internet. Furthermore, many forensic procedures that rely on compression factors become useless when dealing with images that have been posted and downloaded several times since social networking sites frequently auto-compress images (Uppada, Patel, & S, 2023).

There has been Extensive research has focused on the detection of Bangla fake news using textual models (George, Hossain, Bhuiyan, Masum, & Abujar, 2021; Rasel et al., 2022; Wahid, Imran, & Rifat, 2023); however, the exploration of Multimodal Bangla Fake News Detection remains largely unexplored. Addressing this gap, this paper aims to enhance the precision of fake news classification by investigating both unimodal and multimodal techniques. Notably, Bangla presents challenges as a under-Resource language, complicating detection efforts. To overcome these challenges, our research introduces the "MultiBanFakeDetect" dataset, which integrates textual and visual information. We evaluate unimodal methods for text-based fake news detection and fake image detection, alongside exploring various fusion techniques to leverage the synergies between different modalities for improved accuracy in detecting fake news. This comprehensive approach seeks to advance the field by bridging the gap between textual and visual cues in the context of Bangla fake news detection.

The paper offers several key contributions, outlined as follows:

- We curated a diverse dataset named **"MultiBanFakeDetect"**, comprising 9600 text–image pairs sourced from various platforms such as online forums, news websites, and social media networks. This dataset covers a wide spectrum of topics including political, social, technology, sports, and entertainment-related themes. The balanced representation of real and fake instances provides a comprehensive view of contemporary digital content, establishing a solid baseline for accurate comparison and analysis.
- We proposed the **"TextFakeNet"** framework, utilizing state-of-the-art pre-trained language models such as mBERT, XLM-RoBERTa, and DistilBERT architectures. This framework aims to enhance text-based fake news detection by fine-tuning these models on our curated Bangla dataset, MultiBanFakeDetect. By leveraging their advanced semantic and contextual understanding capabilities, TextFakeNet significantly improves the accuracy in distinguishing between genuine and fake news articles in the Bangla language context.
- We developed the **"MultiFusionFake"** framework, an early fusion approach that integrates textual and visual modalities using **DenseNet 169 and mBERT**. MultiFusionFake aims to synergistically combine the strengths of text and image data, demonstrating enhanced performance in the binary classification of fake (1) and non-fake (0) news instances within the Bangla language context. The performance of MultiFusionFake is compared with multiple early, late, and intermediate fusion strategies, highlighting its effectiveness over alternative approaches.
- We extended our approach to multiclass classification, categorizing news instances into four classes: **clickbait, misinformation, rumor, and non-fake**. We developed models using early, late, and intermediate fusion strategies to effectively combine textual and visual features. Our comparative analysis demonstrates how different fusion techniques impact classification performance, providing insights into the most effective approach for distinguishing these categories.
- To gain a deeper understanding of our system's performance in both **binary** (fake vs. non-fake) and **multiclass** (clickbait, misinformation, rumor, and non-fake) classification, we conducted a comprehensive **error analysis** of the Multimodal Bangla Fake News Detection system. For binary classification, our analysis revealed challenges such as misleading text resembling factual content, visually deceptive images, and inconsistencies between textual and visual modalities. In multiclass classification, we identified overlapping characteristics between misinformation and rumors, visually ambiguous clickbait content, and subtle variations in misleading narratives. Additionally, certain cases exhibited discrepancies between text and image cues, leading to misclassifications. These insights guided us in refining our models, improving their robustness, and enhancing their ability to distinguish between different types of fake news.

The remainder of this paper is organized as follows: Section 2 offers an extensive review of related literature, establishing the foundation for our research. Section 3 explores background studies relevant to the field, providing context and supporting information. Section 4 details the dataset creation process used in the study, outlining how the data was collected, processed, and prepared for analysis. Section 5 describes the proposed methodology for multimodal Bangla fake news detection, explaining the techniques and approaches employed. Section 6 interprets the experiments and result analysis. Section 7 examines the errors encountered during the detection process, categorizing misclassifications and identifying areas for improvement. Section 8 discusses practical implementations of the proposed system in combating misinformation across digital platforms. Section 9 examines the study's limitations, discussing potential constraints and areas for improvement. Section 10 proposes directions for future research, suggesting avenues for further investigation and development. Finally, Section 11 provides the conclusion of the research, summarizing the key points and contributions of the study.

2. Related works

The landscape of fake news detection has witnessed significant advancements, driven by both unimodal and multimodal approaches. Unimodal methods primarily leverage textual or visual content to identify misinformation, employing various machine learning and deep learning techniques. In contrast, multimodal approaches integrate data from multiple sources such as text, images, and videos to enhance detection accuracy. This section provides an overview of recent developments in both unimodal and multimodal fake news detection strategies. Tables 1 and 2 provide summaries of the related works.

2.1. Unimodal based fake news detection

The investigation of different machine learning and deep learning models for identifying bogus news in Bangla, as explored by Hossain et al. (2021), utilizes a substantial dataset of 57,000 Bangla news articles. This study employs both traditional methods like Random Forest and Support Vector Machines and advanced models such as Bi-LSTM with GloVe and FastText embeddings, highlighting the superior performance of the Bi-LSTM model with a 96% accuracy rate. Similarly, another research by Hussain et al. (2020a) delves into the effectiveness of machine learning techniques for detecting false news in Bangla. This research employs feature extraction methods like CountVectorizer and TF-IDF Vectorizer, along with supervised algorithms, including Multinomial Naive Bayes (MNB) and Support

Table 1
Comprehensive overview of research on unimodal approaches for fake news detection, emphasizing key methods, techniques, and results.

Modality	Author	Year	Models employed	Key findings
UNIMODAL	Hussain et al. (2020a)	2020	Multinomial Naive Bayes, SVM	In the task of identifying Bangla Fake News, the SVM with a linear kernel achieved the highest accuracy of 96.64%, outperforming the MNB model.
	Hossain, Rahman, Islam, and Kar (2020)	2020	BERT, LSTM	When it came to identifying fake news in Bengali, the BERT model outperformed the LSTM, CNN, SVM, and Naive Bayes models, achieving the highest precision rate of 95%.
	Hossain et al. (2021)	2021	Bi-LSTM, Random forest, SVM and GRU	Achieving a maximum accuracy of 96%, the Bi-LSTM model with GloVe embeddings proved to be effective in handling the complexity of the Bangla language.
	Muzakker, Awosaf, Prottoy, Alvy, and Morol (2022)	2022	Baseline model	Implementing SMOTE improved the F1-score to 93.1%, significantly enhancing the model's performance compared to the baseline.
	Saha, Sarker, Chakraborty, and Yousuf (2022)	2022	BERT, LSTM	In identifying fake news in Bengali, the BERT model outperformed the LSTM, CNN, SVM, and Naive Bayes models, achieving the highest precision rate of 95%.
	Jessica Hauschild (2024)	2024	BERT, Word2Vec, Bag of Words	Compared various word embedding techniques for fake news detection, showing that contextual embeddings significantly enhance classification performance.
	Blackledge and Atapour-Abarghouei (2021)	2022	Transformer-based models	Explored transformers for fake news classification with a focus on generalization, achieving a 10.1% improvement using a two-step pipeline for filtering opinion-based articles.
	Jwa, Oh, Park, Kang, and Lim (2019)	2019	BERT	Demonstrated that analyzing headline-body relationships with deep contextual models improves fake news classification, achieving a 0.14 F-score improvement over previous methods.
	Vaibhav et al. (2019)	2019	Graph neural networks	Proposed a GNN-based model for fine-grained fake news classification, outperforming traditional neural baselines in both in-domain and out-of-domain settings.
	Hosseini, Sabet, He, and Aguiar (2023)	2022	Variational autoencoder, Bi-LSTM	Developed an interpretable fake news detection model by integrating semantic topic modeling with neural embeddings, achieving competitive accuracy with enhanced interpretability.

Vector Machine (SVM) with linear kernels. Notably, the linear kernel SVM classifier achieves a highest accuracy of 96.64%, outperforming the MNB classifier. Further, another study by Muzakker et al. (2022) addresses the issue of unbalanced datasets in the context of identifying false information in Bangla. They employ data modification strategies like SMOTE and layered generalization to enhance model performance on the highly skewed BanFakeNews dataset. Through these techniques, they achieve a 79.1% F1-score using stacking generalization and a 93.1% F1-score via SMOTE, demonstrating efficient methods for mitigating dataset imbalances. Additionally, Saha et al. (2022) investigate a sophisticated approach utilizing the BERT model, enhanced with transfer learning, to identify fake news in Bengali. This model, through the innovative application of Masked Language Modeling (MLM), surpasses other models such as LSTM, SVM, CNN, and Naive Bayes, achieving a remarkable 95% accuracy rate. This underscores the BERT model's efficacy in handling Bengali text and preventing the spread of false information. Another significant effort by Hossain et al. (2020) presents a significant effort to tackle the issue of fake news in Bangla, a language underrepresented in computational research. Their work introduces a large dataset of approximately 50,000 news articles to train advanced NLP models for fake news detection. By examining neural network-based techniques and traditional linguistic features, including pre-trained Transformers models, this research highlights the challenges and advancements in fake news detection in under-resourced languages. Jessica Hauschild (2024) explored word embedding techniques for fake news classification, highlighting contextual word relationships. Building on this, Blackledge and Atapour-Abarghouei (2021) examined transformer generalization across datasets, proposing a two-step filtering pipeline. Similarly, Jwa et al. (2019) leveraged BERT to analyze headline-body relationships, improving F-score by 0.14. Subsequently, Hosseini et al. (2023) developed a deep probabilistic model integrating topic-related features. Al-Tarawneh, Al-irr, Al-Maaitah, Kanj, and Aly (2024) conducted a study comparing TF-IDF, Word2Vec, and FastText embeddings for fake news detection. The results indicated that TF-IDF combined with SVM achieved the highest accuracy at 99.03%, while, in contrast, the MLP model performed best with Word2Vec and FastText embeddings. In general, TF-IDF outperformed the other embeddings across most models, highlighting its superiority in the task of fake news detection. Ali, Ghaleb, Al-Rimy, Alsolami, and Khan (2022) proposed a deep ensemble fake news detection model, achieving a 2.41% increase in F1-score on the challenging LIAR dataset and 100% accuracy on the ISOT dataset, demonstrating the superiority of traditional feature extraction over text embedding techniques. Similarly, Aslam et al. (2021) developed an ensemble-based deep learning model for fake news detection using the LIAR dataset, combining Bi-LSTM-GRU for textual attributes and dense deep learning for other features, achieving an accuracy of 0.898 with high recall and precision, outperforming previous studies. Meanwhile, Abishai Joy and Spezzano (2022) introduced a novel approach to model fake news sharing on social media, incorporating user, network, and news-based features, and achieved an AUROC of 0.97 and an average precision of 0.88, significantly outperforming baseline models and highlighting the effectiveness of news-based features in predicting fake news sharing. In a similar vein, Vaibhav et al. (2019) proposed a graph neural network-based model to capture sentence interactions for fine-grained fake news classification. Their model not only outperformed strong neural baselines but also achieved state-of-the-art accuracy. Furthermore, the model demonstrated impressive generalizability in out-of-domain evaluations, reinforcing its potential for use in a wide range of scenarios. On the other hand, Truică and Apostol (2023) achieved state-of-the-art performance in fake news detection by leveraging document embeddings generated with BART. Both their BiGRU and BiLSTM models reached an accuracy of 99.36% across multiple datasets. The study emphasized that BART-based embeddings consistently outperformed other approaches, regardless of the classifier used, proving the robustness of BART in this domain. Finally, Malliga Subramanian, Shanmugavadivel, Pandiyan, Palani, and Chakravarthi (2025) evaluate Malayalam fake-news detection, which includes binary classification (Fake vs. Original) and five-way categorization using a fact-checked dataset. XLM-RoBERTa led

Table 2
Overview of research studies on multimodal approaches for detecting fake news, highlighting key methodologies, techniques, and findings.

Modality	Author	Year	Models employed	Key findings
MULTIMODAL	Giachanou, Zhang, and Rosso (2020)	2020	Neural networks	Combining textual, visual, and contextual information significantly improves the precision of fake news detection systems. This multimodal approach leverages the strengths of each data type to provide a more comprehensive analysis, resulting in higher accuracy.
	Hossain et al. (2021)	2022	Text: BERT and Images: Swin transformer and VGG19, followed by a fusion strategy	The research introduces the Multimodal Progressive Fusion Network (MPFN) model, which integrates textual and visual features through progressive fusion. The MPFN model demonstrates superior performance in detecting fake news, outperforming existing models across various platforms.
	Segura-Bedmar and Alonso-Bartolome (2022)	2022	CNN, BiLSTM, BERT	Combining text and image data using CNNs, BiLSTMs, and BERT significantly enhances fake news detection accuracy. The integration of these models leverages the strengths of both textual and visual data, leading to more robust detection capabilities.
	Jing, Wu, Sun, Fang, and Zhang (2023)	2022	BERT, VGG19	The integration of external knowledge and user interactions via a knowledge graph enhances detection capabilities, achieving high accuracy scores
	Wang, Mao, and Li (2022)	2022	Fine-Grained Multimodal Fusion Networks (FMFN)	Fine-grained fusion of text and image features using scaled dot-product attention results in superior performance compared to simple concatenation-based methods.
	Guo, Ge, and Li (2023)	2023	Multimodal bilinear pooling, attention mechanism	Multimodal bilinear pooling with an attention mechanism substantially outperforms unimodal methods in accuracy. This approach effectively integrates textual and visual information, leveraging the strengths of both data types. The attention mechanism enhances the interaction between different modalities, leading to more precise detection results.
	Fu and Liu (2023)	2023	Weighted graph convolutional network and a knowledge graph	The integration of external knowledge and user interactions via a knowledge graph enhances detection capabilities. This approach leads to higher accuracy scores in fake news detection. Leveraging knowledge graphs provides a more comprehensive and robust analysis.
	Kalra, Sharma, Vyas, and Chauhan (2023)	2023	ViT, Distil, RoBERTa	Combining textual, visual, and contextual information substantially improves the precision of fake news detection. This multimodal approach leverages the strengths of each data type for more comprehensive analysis.

all submissions, achieving macro–F1 scores of 89.8% on the binary task and 68.2% on the multiclass task. Inspired by these works, our study enhances multimodal Bangla fake news detection by fusing textual and visual cues for improved accuracy and interpretability.

2.2. Multimodal based fake news detection

Fei Wu, Gao, Ji, and Jing (2025) propose BMLHF, which employs a Multi–modal Information Balancing module to dynamically weight semantic and pattern views of text and image. A two-stage Hierarchical Fusion module applies intra-modal multi-view attention followed by inter-modal cross-attention transformers to integrate features. Experiments on Twitter, Weibo and Fakeddit show that BMLHF outperforms state-of-the-art multi–modal fake-news detectors. Consequently, This study (Segura-Bedmar & Alonso-Bartolome, 2022) investigates the effectiveness of unimodal and multimodal methods in identifying false news using deep learning architectures like CNNs, BiLSTMs, and BERT on the Fakeddit dataset. It highlights that a multimodal approach, combining text and visual data, outperforms text-only models. The multimodal CNN model achieved an 87% accuracy rate, demonstrating the importance of integrating visual and textual data to enhance fake news detection. This approach improves both recall and precision across various false news categories. On the other hand, another research by Guo et al. (2023) introduces a novel two-branch network that integrates text and image data to detect fake news, using multimodal bilinear pooling and an attention mechanism. Experiments on Weibo and Twitter datasets show that their model outperforms state-of-the-art methods in accuracy, precision, and recall, demonstrating the potential of deep learning for reliable fake news detection across platforms. Moreover, another research (Jing et al., 2023) introduces the MPFN model, which employs an innovative approach to recognize fake news by gradually integrating multimodal information from textual and visual data. The model combines Swin Transformer and VGG19 for thorough visual analysis and uses a pre-trained BERT for text feature extraction. Results show that MPFN achieves higher accuracy on both Weibo and Twitter datasets compared to other state-of-the-art methods. Specifically, it achieved 83.3% accuracy on Twitter, at least 4.3% higher than other methods, due to its ability to proficiently combine multimodal characteristics for a refined understanding of news authenticity. Additionally, the study by Fu and Liu (2023) presents the A-KWGCN model, a cutting-edge method for identifying fake news by combining user interaction data with outside knowledge. This multimodal approach integrates a weighted graph convolutional network and a knowledge graph to assess content and social interactions effectively. The model achieves high accuracy on Twitter datasets (0.905 and 0.930) by linking news content to related knowledge entities and analyzing user interactions. Ablation studies confirm the importance of the knowledge and weighted GCN modules in enhancing detection. Validated on Twitter15 and Twitter16 datasets, the A-KWGCN model outperforms six comparison models in four assessment indexes. Consequently, Hangloo and Arora (2022) propose a method for multimodal false news detection, integrating knowledge graphs with text and image modalities using a model called KG-GNN (Knowledge Graph-Graph Neural Network). The KG-GNN outperforms baseline techniques in accuracy and resilience, highlighting its effectiveness in contextual and factual verification for fake news detection across platforms. Another study (Giachanou et al., 2020) introduces a novel method for detecting fake news using multimodal inputs—semantic, visual, and textual data are combined to enhance accuracy. The model uses neural networks to analyze post content, related images, and semantic alignment between text and image. Experimental results across three real-world datasets demonstrate superior performance over single-modal systems, emphasizing the importance of text–image congruence in improving detection accuracy and suggesting future enhancements for multimodal information systems. Nevertheless, another research by Hossain et al. (2021) introduces MML (Multi-modal Multi-task Learning), a novel approach that integrates text and image data with external evidence for detecting fake news. Using a multi-level attention mechanism, MML enhances detection capabilities beyond state-of-the-art models by effectively learning relationships between images, texts, and external evidence. Extensive testing on a public dataset shows significant improvements in accuracy and reliability, highlighting the effectiveness of multi-task learning in this domain. Additionally, “FakeRevealer”, developed by Kalra et al. (2023), is a multimodal system that integrates Vision Transformers for visual content and DistilRoBERTa for textual analysis. Using a cosine similarity metric to merge features, FakeRevealer achieves an accuracy of 80% on a Twitter dataset, outperforming previous state-of-the-art techniques by 2.23%. This highlights its effectiveness in leveraging deep learning to understand complex interactions across different data modalities in detecting misleading information. However, another study (Wang et al., 2022) introduces FMFN (Fine-Grained Multimodal Fusion Networks), a novel approach for detecting false news that integrates textual and visual data effectively on microblog platforms. FMFN employs scaled dot-product attention for fine-grained fusion, contrasting with traditional methods that simply concatenate text and image information. Leveraging RoBERTa for contextualized word embeddings and CNNs for visual feature extraction, the model achieves superior performance on a Weibo dataset, demonstrating its ability to accurately identify false information by capturing nuanced interactions between text and image data. Notably, Guo et al. (2025) build AMG, the first fine-grained multimodal fake-news attribution dataset classifying fakes by causes such as image fabrication, non-evidential images, entity inconsistency, event inconsistency and time inconsistency. They introduce MGCA, a multi-granular clue alignment model that extracts and aligns multi-view textual and visual features to jointly detect and attribute fake news. Experiments on AMG reveal its challenge and validate MGCA as a strong baseline. Finally, Yu, Sheng, Lu, Luo, and Zhou (2025) introduce RaCMC, a residual-aware compensation network that fuses cross-modal features via a multiscale compensation module and sharpens real vs. fake distinctions with multi-granularity constraints and interaction exclusivity. Evaluated on Weibo17, Politifact and GossipCop, RaCMC achieved state-of-the-art fake-news detection.

3. Background study

3.1. Fake news detection (text based) models

In the realm of multilingual natural language processing (NLP), the challenge of detecting fake news across diverse languages has spurred the development of sophisticated pre-trained language models (PLMs). These models, leveraging extensive training on vast multilingual datasets, have demonstrated remarkable efficacy in understanding and processing text in numerous languages, including Bangla. Two prominent models in this domain, mBERT (Multilingual BERT) and XLM-RoBERTa, have emerged as powerful tools for addressing the intricacies of fake news detection. Their architectures, rooted in advanced Transformer-based mechanisms, enable these models to navigate the complexities of language semantics, grammar, and context with impressive accuracy. Below, we delve into the distinctive features and capabilities of mBERT and XLM-RoBERTa, highlighting their roles in enhancing the reliability of Bangla Fake News Detection.

3.1.1. mBERT (multilingual BERT)

mBERT (Devlin, Chang, Lee, & Toutanova, 2018) is a powerful pre-trained language model (PLM) designed for Bangla Fake News Detection, offering a versatile approach to understanding and processing text in a remarkable 104 languages. Rooted in the renowned BERT architecture, mBERT sets itself apart through its extensive training on vast quantities of text data sourced from Wikipedia articles across these languages. This comprehensive training equips mBERT with the capability to discern the intricacies of Bangla language, encompassing its vocabulary, grammar, and semantics. The pre-training phase of mBERT is where its true power shines. During this phase, the model is exposed to an immense dataset of multilingual text, devoid of any specific task labels. The core mechanisms driving mBERT’s pre-training process include Masked Language Modeling (MLM) and Sentence Order Prediction. In MLM, mBERT tackles the challenge of predicting masked words within a sentence based on contextual cues, facilitating a deeper understanding of word relationships within the Bangla language. Meanwhile, the Sentence Order Prediction task enhances mBERT’s comprehension of sentence structure and coherence across languages, further bolstering its effectiveness in Bangla Fake News Detection.

3.1.2. XLM-RoBERTa

XLM-RoBERTa (Conneau et al., 2019) stands out as a formidable tool for Fake News Detection, leveraging its prowess in multilingual natural language processing (NLP) to effectively sift through text data across a vast spectrum of 100 languages. Through its extensive pre-training regimen on a diverse dataset spanning multiple languages, XLM-RoBERTa develops a nuanced understanding of language intricacies, encompassing vocabulary, grammar, and contextual word relationships. This understanding is further enhanced by its dual learning objectives during pre-training: Masked Language Modeling (MLM) and Sentence Order Prediction, which fortify its grasp on word relationships and sentence coherence. At its core, XLM-RoBERTa’s Transformer encoder, equipped with self- attention mechanisms, facilitates the analysis of complex relationships within sentences, enabling robust detection of deceptive content regardless of the language used. With the flexibility to fine-tune for specific tasks like Bangla Fake News Detection by adding task-specific output layers, XLM- RoBERTa emerges as a powerful ally in combating misinformation across linguistic boundaries.

3.1.3. DistilBERT

DistilBERT (Faria et al., 2024) is a lightweight pre-trained language model (PLM) optimized for efficiency while retaining strong language understanding, making it suitable for Bangla Fake News Detection. Derived from BERT via knowledge distillation, it achieves faster processing with fewer parameters while preserving performance. During pre-training, DistilBERT learns linguistic structures through Masked Language Modeling (MLM) but omits the Sentence Order Prediction task, ensuring efficient learning. Its Transformer-based architecture, enhanced by self-attention, enables effective detection of deceptive content. With fine-tuning, DistilBERT offers a powerful yet computationally efficient solution for misinformation detection.

3.2. Fake news detection (image based) models

To address the complexities of fake news detection through images, advanced deep learning architectures have been employed. These architectures leverage the power of convolutional neural networks (CNNs) to analyze and interpret visual data, thereby enabling the identification of misleading or deceptive imagery. Among the prominent CNN architectures used in this domain are ResNet, DenseNet, and MobileNet, each offering unique advantages in terms of depth, connectivity, and efficiency. The following sections delve into the specifics of these architectures, highlighting their design principles, innovations, and contributions to the field of image-based fake news detection.

3.2.1. ResNet architectures

ResNet 50 (He, Zhang, Ren, & Sun, 2016) is a convolutional neural network architecture renowned for its effectiveness in tackling the challenge of training very deep networks. It introduces residual learning with 50 layers, addressing the vanishing gradient problem by learning residuals relative to the input. This is achieved through residual connections that facilitate smoother information flow across the network, thereby enhancing training efficiency and improving performance across a variety of computer vision tasks. ResNet 101 builds upon the success of ResNet 50 by extending its depth to 101 layers. This deeper architecture further enhances the model's capacity for learning intricate features and representations in data. ResNet 101 (He et al., 2016) continues to utilize residual connections to maintain effective gradient flow, enabling more effective training and boosting overall performance. Extending the depth even further, ResNet 152 (He et al., 2016) pushes the boundaries with 152 layers. This advancement allows for a more sophisticated hierarchy of feature learning, facilitating the extraction of detailed patterns across multiple levels of abstraction. The additional layers empower the network to discern subtle details and make informed decisions during inference, particularly benefiting applications that require complex feature extraction and semantic understanding from input data.

3.2.2. DenseNet architectures

DenseNet 121 (Huang, Liu, Van Der Maaten, & Weinberger, 2017), an instantiation of the DenseNet architecture, revolutionizes neural network design by establishing dense connections between all layers in a feed-forward manner. Unlike traditional architectures, where information flows sequentially layer by layer, DenseNet 121 allows each layer to receive inputs from all preceding layers, promoting dense connectivity throughout the network. This innovation enables unparalleled feature reuse and mitigates the vanishing-gradient problem by ensuring direct gradient propagation across layers during backpropagation. Building upon this foundation, DenseNet 169 (Huang et al., 2017) extends the depth to 169 layers, further enhancing its capacity for feature extraction and representation learning while maintaining dense connectivity. This architecture fosters efficient information exchange and extensive feature reuse, facilitating the capture of intricate patterns and semantics in data. DenseNet 169 excels in various computer vision tasks such as image classification, segmentation, and object detection, contributing significantly to advancements in the field. DenseNet 201 (Huang et al., 2017) continues this evolution by increasing the depth to 201 layers. This extension allows for even more comprehensive feature extraction and representation learning capabilities. DenseNet 201 retains the dense connectivity characteristic of its predecessors, ensuring robust gradient flow and promoting effective learning across the network.

3.2.3. MobileNet architectures

MobileNet V2 (Howard et al., 2017) is a specialized convolutional neural network architecture designed for efficient image classification on mobile and embedded devices, emphasizing computational efficiency without sacrificing accuracy. It achieves this through innovative techniques such as depthwise separable convolutions, which significantly reduce computational costs by applying filters to individual input channels followed by pointwise convolutions for feature combination. MobileNet V2 also incorporates linear bottlenecks to further streamline complexity and utilizes residual connections inspired by ResNet architectures to enhance gradient flow and overall performance. Its core building blocks, including inverted residual blocks, leverage these efficient strategies for lightweight yet effective image data processing, supported by standard convolution layers for initial feature extraction and final classification tasks. MobileNet V3 (Howard et al., 2019) builds upon MobileNet V2's foundations with enhancements like the H-Swish activation function for reduced computational overhead and optional Squeeze-and-Excite (SE) blocks to enhance feature representation through dynamic channel weight adjustments. MobileNet V3's balanced scaling of network width and depth, coupled with its refined building blocks, establishes it as a benchmark for efficient image classification on mobile and embedded platforms, setting new standards for performance and resource utilization.

3.3. Evaluation metrics for fake news detection

3.3.1. Accuracy

Accuracy (Powers, 2020) is a crucial metric in Fake News detection that measures the overall correctness of the model. It is the ratio of correctly predicted instances (both true positives and true negatives) to the total number of instances. High accuracy indicates that the model performs well in distinguishing between fake and non-fake news.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

3.3.2. Precision

Precision (Powers, 2020) measures the accuracy of the positive predictions, indicating how many of the instances identified as fake news are actually fake news. It is calculated as the ratio of true positives to the sum of true positives and false positives. High precision means the model makes fewer false positive errors.

$$\text{Precision}_{\text{binary}} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Precision}_{\text{weighted}} = \sum_{c \in C} w_c \frac{TP_c}{TP_c + FP_c} \tag{3}$$

3.3.3. Recall

Recall (Powers, 2020), also known as sensitivity or true positive rate, measures the ability of the model to correctly identify all actual fake news instances. It is the ratio of true positives to the sum of true positives and false negatives. High recall indicates that the model successfully captures most of the fake news.

$$\text{Recall}_{\text{binary}} = \frac{TP}{TP + FN} \tag{4}$$

$$\text{Recall}_{\text{weighted}} = \sum_{c \in C} w_c \frac{TP_c}{TP_c + FN_c} \tag{5}$$

3.3.4. F1 score

The F1 score (Powers, 2020) is the harmonic mean of precision and recall, providing a single metric that balances both. It is particularly useful when there is an uneven class distribution or when both precision and recall are important. A high F1 score means the model has a good balance of precision and recall, effectively identifying fake news while minimizing false positives and false negatives.

$$\text{F1}_{\text{binary}} = \frac{2 \cdot \text{Precision}_{\text{binary}} \cdot \text{Recall}_{\text{binary}}}{\text{Precision}_{\text{binary}} + \text{Recall}_{\text{binary}}} \tag{6}$$

$$\text{F1}_{\text{weighted}} = \sum_{c \in C} w_c \cdot \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}, \quad \text{where} \quad w_c = \frac{N_c}{N} \tag{7}$$

4. Dataset creation

4.1. Data collection

(a) ***Collection Approach*:** In our inclusive Bangla fake news collection, we gathered both real and fake data across various platforms, including online forums, news websites, Facebook, X (formerly known as Twitter), and Instagram, to ensure a comprehensive overview of digital content. Categories encompassed political, sports, technology, lifestyle, and entertainment, with

Table 3
Characteristics of headlines by category for various news categories.

Category	Total words for headline	Total unique words	Max headline length	Min headline length
Entertainment	5457	2171	13	93
Sports	5181	2141	81	16
Technology	6389	2198	133	12
National	6314	2511	112	18
Lifestyle	5760	2196	122	15
Politics	6602	2294	133	17
Education	6298	2497	140	17
International	5849	2389	88	17
Crime	6147	1905	91	17
Finance	5823	2088	85	15
Business	5887	2163	92	16
Miscellaneous	5400	2453	350	7

each category including misinformation, rumors, and clickbait related to relevant events and figures, covering a wide range of topics from political events to sports events, technological innovations to lifestyle trends, and entertainment industry updates, etc. This approach mirrors the complexity of misinformation dissemination across various subjects and platforms.

(b) ***Balanced Representation***: Throughout the data collection process, we maintained a balanced representation of both real and fake instances to ensure a comprehensive dataset. This allowed for a thorough analysis and model training, capturing the subtleties and complexities of fake news across various contexts. The collected data provided a solid baseline for comparison with real data, facilitating robust evaluation and validation of detection models. By establishing this baseline, we can effectively measure the performance and accuracy of our detection systems, ensuring their reliability in identifying fake news in the Bangla language context.

4.2. Dataset characteristics

Our dataset is specially designed to include both visual and textual content, enhancing both the quantity and quality of data that can be analyzed. Each instance in the collection consists of a text–image pair, where the image serves as a visual representation and the textual component acts as a caption or description. The "headline" and "description" columns in our dataset correspond to its textual component. The textual data in these areas was collected from various internet forums, social media posts, and headlines. The headline provides a brief overview of the major idea or subject matter of the related material, while the description offers more information or background. The "image_id" column in our dataset contains distinct IDs for every image, representing the visual component. These images are sourced from a variety of platforms and include screenshots, infographics, memes, and other visual information. The inclusion of visual material in the dataset is vital for a thorough study, given its importance in the spread and interpretation of false information. Additionally, our dataset is categorized into the following categories: Entertainment, Sports, Technology, National, Lifestyle, Politics, Education, International, Crime, Finance, Business, and Miscellaneous. Each instance is also labeled with one of the following types of fake news: misinformation, rumor, clickbait, or non-fake. The "label" column assigns a value of 0 for non-fake instances and 1 for fake instances. This labeling scheme allows for effective classification and analysis of the dataset.

1. ***Misinformation***: This refers to general false or misleading content that is factually incorrect or inaccurate, regardless of intent. In our classification framework, this class includes misleading content that does not fall specifically under the characteristics of rumors (speculative/unverified) or clickbait (sensationalism for attention), thereby serving as a broader umbrella for general inaccuracy.
2. ***Rumor***: Rumors are speculative or unverified claims that lack substantiated evidence at the time of dissemination. While they may eventually be validated or disproven, they are categorized based on their unconfirmed status and tendency to propagate rapidly, particularly through social networks.
3. ***Clickbait***: This category includes content that uses exaggerated, sensational, or misleading headlines primarily to attract attention and generate clicks. It often sacrifices accuracy for virality, focusing on engagement over truthful reporting.
4. ***Non-fake***: This label is assigned to instances that are verified to be accurate and truthful. These instances are not misleading or false and serve as a control group in the dataset for the purpose of comparison and analysis.

Tables 3, 4, and 5 provides an overview of the dataset characteristics, showcasing the text–image pairs along with associated metadata such as category, types of fake news, and label indicating the authenticity of the content. This structured format enables researchers to efficiently analyze and explore the dataset for various research purposes. Fig. 1 illustrates the step-by-step process of dataset creation, highlighting the essential phases and methodologies used in its development. Figs. 2 and 3 showcase examples of text–image pair visualizations from various news categories in the MultiBanFakeDetect dataset. The dataset used in this research is publicly accessible through: Dataset Link

4.3. Data annotation

Step (1) Annotator Recruitment: The process of annotator recruitment for our Multimodal Bangla Fake News Detection project is meticulously tailored to identify individuals with specialized expertise in discerning fake news within the Bangla language landscape. Our candidates possess a profound understanding of fake news detection principles, honed through active engagement and extensive exposure to social media platforms. We prioritize recruiting individuals with a proven track record of accurately identifying and debunking fake news across various categories, showcasing their nuanced comprehension of misinformation tactics. Annotators are selected based on their adeptness in recognizing linguistic nuances and visual cues indicative of fake content, coupled with their unwavering commitment to truth in online discourse. Furthermore, candidates are categorized according to their domain-specific knowledge to ensure that each annotator brings targeted expertise relevant to specific categories such as politics, entertainment, technology, and more. Our training sessions are meticulously curated to empower annotators with comprehensive insights into category-specific nuances, project objectives, annotation guidelines, and stringent evaluation criteria. Table 6 gives Summary of Annotator Profiles. In assessing the trustworthiness of each annotator, we employ a multi-faceted approach. This includes reviewing their past work and contributions in the field of fake news detection, evaluating their critical thinking skills, attention to detail, and ability to maintain objectivity in assessments. Additionally, we conduct interviews with each candidate to gauge their commitment to upholding ethical standards and adherence to project guidelines. Through this rigorous selection process, we identify annotators who not only demonstrate expertise in fake news detection but also exhibit integrity and professionalism. These individuals undergo training sessions tailored to their specific areas of domain knowledge, ensuring they are well-equipped to accurately evaluate multimodal content across various categories. As a result of this meticulous vetting process, we assemble a team of annotators who are highly skilled, trustworthy, and dedicated to maintaining the rigorous standards of our project. Additionally, annotators are compensated according to Bangladesh's wage regulations to ensure fair compensation for their valuable contributions. Initially, we invited 13 individuals to assist with the annotation process, after which we assessed the suitability of each candidate based on the aforementioned criteria.

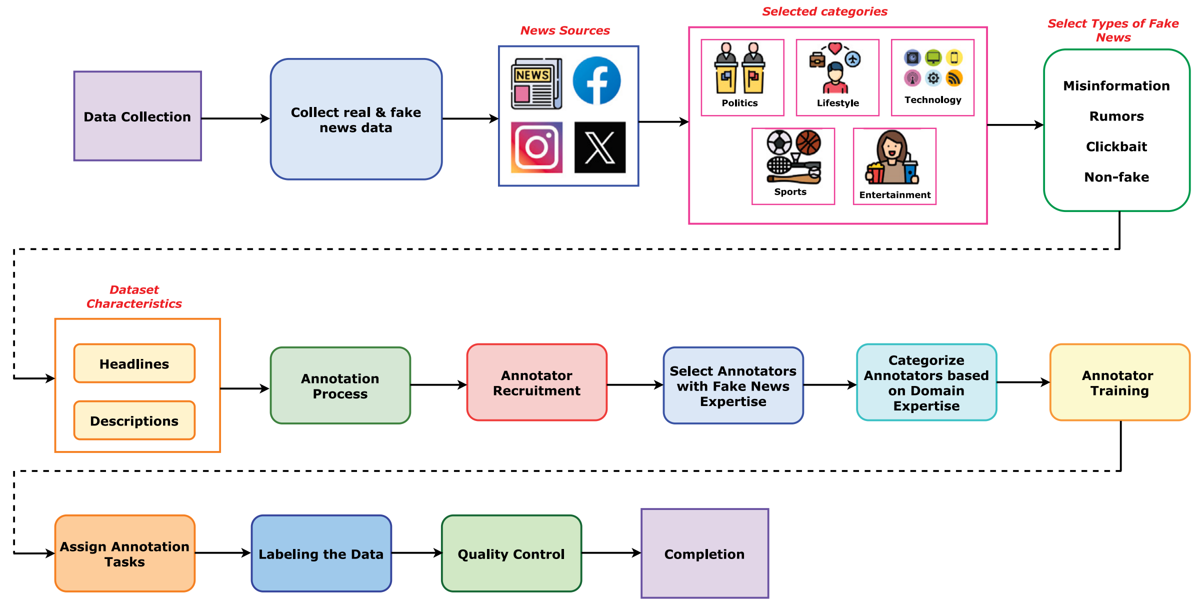

Fig. 1. Detailed step-by-step approach for dataset creation, outlining the key phases and methodologies involved in building the dataset.

Table 4
Characteristics of description by category for various news categories.

Category	Total words for description	Total unique words	Min description length	Max description length
Entertainment	118 954	14 726	2288	337
Sports	106 320	11 715	2673	134
Technology	56 727	9168	939	197
National	64 963	9696	1182	210
Lifestyle	70 819	11 545	1469	192
Politics	93 241	10 253	1850	152
Education	88 087	11 440	1784	49
International	96 210	12 367	1996	114
Crime	68 455	7877	2154	224
Finance	71 387	7888	2028	242
Business	66 200	9581	1933	200
Miscellaneous	88 005	14 240	3721	215

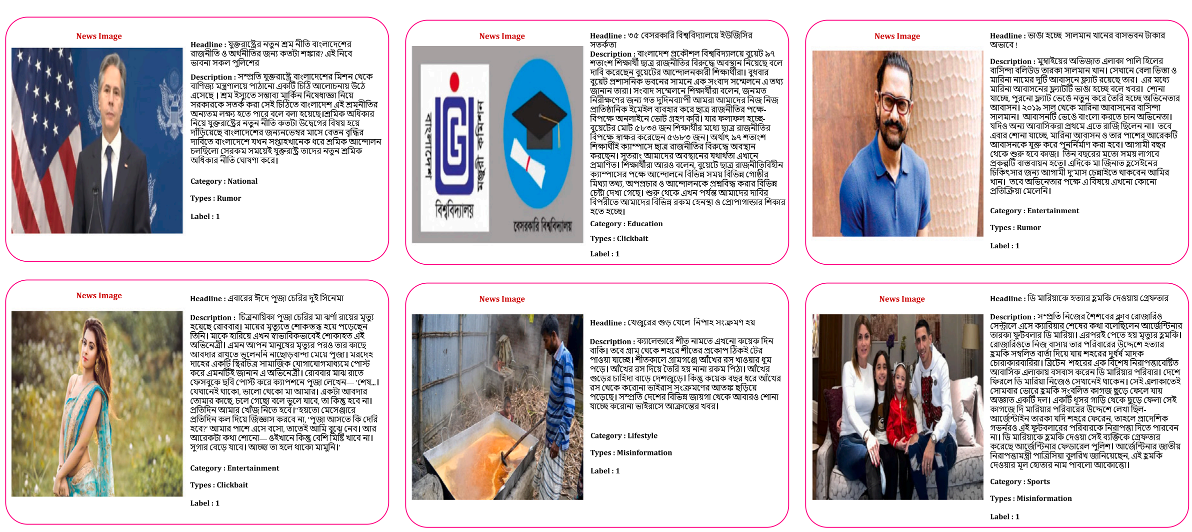

Fig. 2. Representative examples of text–image pair visualizations across different news categories for multiclass classification.

Step (2) Annotation Guideline: We have developed a comprehensive annotation guideline to facilitate the accurate labeling of fake news instances within our dataset. This guideline delineates specific criteria and instructions aimed at discerning between various types of fake news, including misinformation, rumors, and clickbait. First, we address misinformation, which is the intentional spread of incorrect or erroneous information in an effort to trick or mislead readers. Annotators are expected to carefully evaluate textual and visual content,

News Image	headline	headline (English Translation)	description	description (English Translation)	Category	types of fake news	label	Remarks	Remarks (English Translation)
	অবন্তিকার আত্মহত্যায় সহপাঠী ও সহকারী প্রক্টরের খণ্ডিত সংশ্লিষ্টতা পাওয়া গেছে: ডিএমপি	Fragile involvement of classmate and assistant proctor found in Avantika's suicide: DMP	জগন্নাথ বিশ্ববিদ্যালয়ের ছাত্রী ফাইরুজ সাদাফ অবন্তিকার আত্মহত্যার ঘটনায় তাঁর সহপাঠী রায়হান সিদ্দিকী আম্মান ও সহকারী প্রক্টর দ্বীন ইসলামের খণ্ডিত সংশ্লিষ্টতা পাওয়া গেছে বলে জানিয়েছেন ঢাকা মহানগর পুলিশের (ডিএমপি) অতিরিক্ত কমিশনার খ মহিদ উদ্দিন। আজ রোববার দুপুরে এক সংবাদ সম্মেলনে তিনি এ কথা বলেন।	Dhaka Metropolitan Police (DMP) Additional Commissioner K. Mahid Uddin has confirmed that there is evidence of broken ties between Faiyruz Sadaf, a student of Jagannath University, and his accomplice Rayhan Siddiqui Aman and assistant proctor Deen Islam in the case of Avantika's suicide. He stated this at a press conference today.	crime	non-fake	0	এই সংবাদটি প্রথম আলো, দৈনিক নয়া দিগন্ত সহ বিশ্বস্ত বাংলা সংবাদপত্রগুলোতে প্রকাশিত হয়েছে। ঢাকা মহানগর পুলিশের (ডিএমপি) অতিরিক্ত কমিশনার খ মহিদ উদ্দিন নিজেই সংবাদ সম্মেলনে এ তথ্য নিশ্চিত করেছেন। তাই, এই খবরটি সত্য বলে মনে করা হয়। অবন্তিকার আত্মহত্যার ঘটনাটি বেদনাদায়ক এবং দুঃখজনক। তদন্ত চলমান থাকায়, এখনই রায়হান সিদ্দিকী আম্মান ও দ্বীন ইসলামের বিরুদ্ধে চূড়ান্ত মন্তব্য করা উচিত নয়। তদন্তের মাধ্যমে তাদের সম্পৃক্ততার সত্যতা যাচাই করা প্রয়োজন এবং নিরপরাধ প্রমাণিত না হওয়া পর্যন্ত সকলের প্রতি শ্রদ্ধাশীল আচরণ করা উচিত। অবন্তিকার মা তার মেয়ের আত্মহত্যার জন্য রায়হান সিদ্দিকী আম্মান ও দ্বীন ইসলামকে দায়ী করে মামলা করেছেন এবং জগন্নাথ বিশ্ববিদ্যালয় কর্তৃপক্ষ তদন্তে সহযোগিতা করছে।	This news has been published in leading Bengali newspapers such as Prothom Alo and Daily Naya Diganta. Additional Commissioner K. Mahid Uddin himself confirmed this information at the press conference. Therefore, this news is considered true. Avantika's suicide incident is tragic and sorrowful. Investigation is ongoing, and it is inappropriate to make conclusive statements against Rayhan Siddiqui Aman and Deen Islam at this time. It is necessary to verify the truth of their involvement through investigation, and everyone should behave respectfully until their innocence is proven. Avantika's mother has filed a case against Rayhan Siddiqui Aman and Deen Islam for her daughter's suicide, and Jagannath University authorities are cooperating in the investigation.
	রাশমিকা মানদানার ভুয়া ভিডিও তৈরিতে ব্যবহার হয়েছে যে ডিপফেক প্রযুক্তি	Deep fake technology used to create fake video of Rashmika Mandana	মোবাইল ফোন চার্জ করতে দিয়ে অনেকেই কানে লাগিয়ে কথা বলেন বা গেম খেলেন বা ইন্টারনেট ব্রাউজ করেন। কিন্তু চার্জ দেওয়ার সময়ে ফোন ব্যবহার করাটা কি আদৌ নিরাপদ? প্রযুক্তিবিদরা বলছেন, ভাল ব্র্যান্ডের মোবাইল ফোন আর ওই একই কোম্পানির চার্জার ব্যবহার করলে বিপদের সম্ভাবনা অনেকটাই কম।অনেক ঘটনার কথাই জানা যায়, যেখানে মোবাইল ফোন বিস্ফোরণে কারও মৃত্যু হয়েছে অথবা কারও শরীরের কোনও অংশ ঝলসে গেছে।গত সপ্তাহে এরকমই এক ঘটনা ঘটেছে পশ্চিমবঙ্গের দুর্গাপুর শহরে। রিয়া ব্যানার্জী নামে ২২ বছরের এক তরুণী মোবাইলে চার্জ দেওয়ার সময়েই কথা বলছিলেন। হঠাৎই বিস্ফোরণ ঘটে। অগ্নিদগ্ধ অবস্থায় হাসপাতালে নিয়ে গেলে তার মৃত্যু হয়।	Many people talk or play games or browse the internet while charging their mobile phones. But is it safe to use the phone while charging? Technicians say that using a good brand of mobile phone and the charger of the same company is much less dangerous. Many cases are known, where someone died or some part of the body was burnt due to mobile phone explosion. Last week, one such incident happened in Durgapur city of West Bengal. A 22-year-old girl named Rhea Banerjee was talking while charging her mobile phone. Suddenly there was an explosion. He died after being rushed to the hospital.	technology	clickbait	1	এই শিরোনামটি স্পষ্টতই ক্লিকবেইট হিসেবে ব্যবহার করা হচ্ছে, তবে এটি রাশমিকা মানদানার ভুয়া ভিডিও সম্পর্কে দাবি করার সময় একটি ঘটনার নিয়ে বিস্তারিত আলোচনা করেনি। বরং এটি মোবাইল ফোন চার্জ করার সময় বিস্ফোরণের ঘটনার উল্লেখ রয়েছে। এই দুটি ঘটনার মধ্যে কোন সম্পর্ক নেই। পাঠকদের মনোযোগ আকর্ষণ করার জন্য শিরোনাম ও বিবরণ পড়া উচিত এবং বিশ্বস্ত উৎস থেকে সংবাদ সংগ্রহ করা গুরুত্বপূর্ণ। অতিরিক্তভাবে, মোবাইল ফোন চার্জ করার সময় সতর্কতা অবলম্বন করা জরুরি এবং ভাল ব্র্যান্ডের চার্জার ব্যবহার করা উচিত।	The headline is obviously being used as clickbait, but it doesn't elaborate on an incident when Rashmika Mandana's fake video was claimed. Rather it refers to incidents of explosions while charging mobile phones. There is no connection between these two events. Headlines and descriptions should be read to grab readers' attention, and it's important to gather news from trusted sources. Additionally, it is important to be careful while charging the mobile phone and use a good brand of charger.
	নাম থেকে 'খান' কেটে দেওয়ার কারণ জানালেন মিথিলা	Mithila explained the reason for cutting off 'Khan' from the name	রাফিয়াত রশিদ সারা দুই বাংলার জনপ্রিয় অভিনেত্রী। যাকে মিথিলা নামেই দেশের মানুষ বেশি চেনেন। কিন্তু এই অভিনেত্রীর নামের মাঝে 'খান' রয়েছে। যেটা তিনি নিজ নাম থেকে বাদ দিয়েছেন। তবে কেন এমনটা করেছেন ওপার বাংলার স্বনামধন্য পরিচালক সৃজিতের স্ত্রী? সম্প্রতি এক সাক্ষাৎকারে সে কারণটাই জানিয়েছেন তিনি। সারার ভাষায়- হ্যাঁ, আমার নামে 'খান' আছে, তবে সেটা ব্যবহার করি না। আসলে আমার আব্বু আর দাদার নামে খান আছে। এত বড় নাম রাখলে খুব লম্বা হয়ে যেত ব্যাপারটা। তাই এসএসসি রেজিস্ট্রেশনের সময় খান শব্দটা বাদ দিয়েছি।জুক্ত করেছি অধিকারি। সারা বিয়ে করেছিলেন দেশের জনপ্রিয় অভিনেতা ও সংগীতশিল্পী তাহসান খানকে। অনেকেই মনে করেন, তাদের বিচ্ছেদের পরেই নাম থেকে খান বাদ দিয়েছেন অভিনেত্রী। কিন্তু তেমনটা নয়, মিথিলার কথায়—নাম থেকে 'খান' বাদ পড়েছে তাহসানকে বিয়ের আগেই।	Rafiat Rashid is a very popular actress of Bengal. Which is known as Mithila by the people of the country. But this actress has 'Khan' in her name. which he omitted from his name. But why did the wife of the well-known director Srijit of Opar Bengal do this? He said that reason in a recent interview. In Sara's words - Yes, I have 'Khan' in my name, but I don't use it. In fact, I have Khan in the name of my father and grandfather. It would have been too long to keep such a big name. So I omitted the word Khan during SSC registration. I added Adhikari. Sara married the country's popular actor and musician Tahsan Khan. Many people think that the actress dropped Khan from her name after their separation. But not like that, according to Mithila - 'Khan' was dropped from the name before Tahsan got married.	entertainment	rumor	1	এই খবরটি যদি সত্য হয়, তবে মিথিলা রশিদের নাম পরিবর্তনের পেছনের কারণে অনেক গুঞ্জন ও তত্ত্বাবধান রয়েছে। অনেকে মনে করেন যে, মিথিলা রশিদ তার সাক্ষাৎকারে প্রকাশিত অবস্থানে থাকার পর তার স্বামী তাহসান খানের সাথে বিচ্ছেদের পর নাম থেকে 'খান' বাদ দিয়েছেন। তবে মিথিলা নিজে এক সক্ষম সাক্ষাৎকারে জানানো সত্যতা, যে তিনি তার এসএসসি রেজিস্ট্রেশনের সময় নামের দৈর্ঘ্য কমানোর জন্য মাত্র 'খান' বাদ দিয়েছিলেন। তার এই বক্তব্যের সত্যতা যাচাই করার জন্য আরও তথ্যপুষ্টি প্রয়োজন। ব্যক্তিগত পছন্দের ভিত্তিতে নাম পরিবর্তন করা একটি ব্যক্তির অধিকার, তবে সেটির পেছনের সত্যতা নিশ্চিত হওয়া গুরুত্বপূর্ণ।	If this news is true, then the reason behind Mithila Rashid's name change has been a lot of buzz and oversight. Many believe that Mithila Rashid dropped the 'Khan' from her name after her split with her husband Tahsan Khan, after the position she revealed in an interview. But the fact that Mithila herself told in an able interview, that she only dropped 'Khan' to reduce the length of her name during her SSC registration. More information is needed to verify the truth of his statement. It is a person's right to change name based on personal choice, but it is important to ascertain the truth behind it.
	চালের বাজারে কারসাজি নেপথ্যে কর্মকর্তারা	Officials behind manipulation in rice market	খাদ্য বিভাগের মাঠ পর্যায়ের কিছু অসাধু কর্মকর্তা-কর্মচারীর দায়িত্ব অবহেলা, গাফিলতি ও লোভের কারণে চালের বাজার স্বাভাবিক রাখা যাচ্ছে না। খাদ্য বিভাগের দেওয়া ফুড গ্রেইন লাইসেন্সের শর্ত ব্যবসায়ীরা মানছে কি না তা দেখার দায়িত্ব তাদের। তারা সরকারি দায়িত্ব সঠিকভাবে পালন করছেন না। তারা চাকরি করেন সরকারি কিন্তু স্বার্থ সংরক্ষণ করেন অসৎ ব্যবসায়ীদের। চালের বাজারে কারসাজির নেপথ্যে এসব অসাধু কর্মকর্তা। তাদের এ ধরনের দ্বিচারিতায় বিব্রত খাদ্য মন্ত্রণালয় ও অধিদপ্তর। ফলে গত ১৮ ফেব্রুয়ারি খাদ্য মন্ত্রণালয় থেকে চালের বস্তায় ধান চালের জাত, মিলের এবং মিলারের নাম, উৎপাদনের তারিখ, ওজন এবং মূল্য লেখা বাধ্যতামূলক করে জারি করা পরিপত্র শতভাগ বাস্তবায়ন নিয়ে সংশয়ে আছে খাদ্য মন্ত্রণালয়।	The rice market cannot be kept normal due to negligence, carelessness and greed of some dishonest officers and employees at the field level of the food department. It is their responsibility to see whether traders are complying with the food grain license conditions given by the food department. They are not performing government duties properly. They work in the government but protect the interests of dishonest businessmen. These dishonest officials are behind the manipulation in the rice market. The food ministry and directorate are embarrassed by their duplicity. As a result, the Ministry of Food is doubtful about the 100% implementation of the circular issued by the Ministry of Food on February 18 making it mandatory to write the variety of rice, miller and miller's name, production date, weight and price on rice sacks.	finance	misinformation	1	এই খবরটি যুগান্তর, প্রথম আলো, এবং বাংলাদেশ প্রতিদিন সহ বিশ্বস্ত বাংলা সংবাদপত্রগুলোতে প্রকাশিত হয়েছে, তবে এর কিছু অংশ যদি সত্য হয়, তবে বিভিন্ন জটিল কারণে চালের বাজারে অস্থিরতা বিরাজ করছে। এই দাবিটি সম্পূর্ণ সত্য বলে মনে করা যায় না, কারণ চালের বাজারের অস্থিরতার জন্য একাধিক পার্থক্যমূলক কারণ থাকতে পারে, যেমন বাজার ব্যবস্থা, সরকারি নীতি, আন্তর্জাতিক বাজারের অবস্থা ইত্যাদি। এটি উপস্থাপন করতে হলে আরও তথ্যপুষ্টি প্রয়োজন, যাতে বিশ্বাস পাওয়া যায় যে, কোনো নিশ্চিত দলিল বা তথ্যের বিশুদ্ধতা পরিশোধিত হয়েছে। সরকার চালের বাজার স্থিতিশীল রাখতে প্রসারিত পদক্ষেপ নিচ্ছে, তবে বাজারে অসৎ ব্যবসায়ীদের বিরুদ্ধে ব্যবস্থা নেওয়া হচ্ছে এবং নজরদারি জোরদার করা হচ্ছে।	This news has been published in trusted Bengali newspapers including Jugantar, Prothom Alo, and Bangladesh Pratidin, but if some part of it is true, there is volatility in the rice market due to various complex reasons. This claim cannot be considered completely true, as there may be multiple differential factors for rice market volatility, such as market systems, government policies, international market conditions, etc. It requires more information to be presented, so that there is confidence that the authenticity of any certified document or information has been verified. The government is taking extensive measures to stabilize the rice market, but action is being taken against unscrupulous traders in the market and surveillance is being strengthened.

Fig. 3. Visualization of text–image pairs data across various fake news types, highlighting key features and remarks explaining why the news is classified as fake.

Table 5
Distribution of fake news types by category across various news categories.

Category	Misinformation	Rumor	Clickbait
Entertainment	130	130	140
Sports	130	130	140
Technology	131	138	131
National	130	130	140
Lifestyle	130	130	140
Politics	140	120	140
Education	140	120	140
International	140	120	140
Crime	140	120	140
Finance	130	130	140
Business	140	120	140
Miscellaneous	130	130	140

Table 6
Summary of annotator profiles involved in the study.

Annotator ID	Qualifications	Experience	Expertise
Annotator 1	Undergraduate	1 year	Computer vision
Annotator 2	Undergraduate	1 year	Multimodal research
Annotator 3	Undergraduate	2 years	Fake data labeling
Annotator 4	Graduate	1 year	Fake data labeling
Annotator 5	Graduate	3 years	Computer vision
Annotator 6	Undergraduate	2 years	Fake data labeling
Annotator 7	Undergraduate	2 years	Multimodal research
Annotator 8	Undergraduate	1 year	Computer vision
Annotator 9	Graduate	2 years	Multimodal research
Annotator 10	Graduate	2 years	Computer vision
Annotator 11	Graduate	1 year	Fake data labeling
Annotator 12	Undergraduate	2 years	Fake data labeling
Annotator 13	Graduate	2 years	Computer vision

closely examining any claims or statements that are clearly false or deceptive. This entails a thorough examination of the factual accuracy of the information presented. Second, our guideline tackles rumors, which are unproven or conjectural statements that become viral on the internet in the absence of supporting data. It is recommended that annotators should be alert in spotting cases when facts are presented as fact even in the absence of reliable sources or verification. This necessitates a keen eye for discerning between verified information and unsubstantiated claims. Finally, we discuss clickbait, which is defined as content that uses dramatic headlines or thumbnails to draw readers in and get them to click. The responsibility of annotators is to assess

Table 7
Annotation quality metrics for types of fake news.

Class	Cohen's Kappa	Fleiss Kappa
Misinformation	0.83	0.85
Rumor	0.81	0.88
Clickbait	0.86	0.82
Non-fake	0.92	0.94
Overall	**0.85**	**0.87**

the usage of attention-grabbing strategies used to persuade visitors to click on the material, such as false claims, inflammatory language, or deceptive images.

Step (3) Annotation Process: Our annotation process starts with in-depth training sessions for annotators, with the goal of acquainting them with the annotation standards and criteria. By giving annotators a range of examples of both fake and real news, we enable them to get a sophisticated comprehension of the traits that set false news apart. Annotators may mark instances according to established criteria owing to our carefully thought-out annotation interface, which makes the annotation process easier. This interface includes features such as the ability to highlight Bangla text passages, mark images, and provide comments or explanations. We periodically perform inter-annotator agreement tests to guarantee consistency and dependability. A selection of cases is labeled separately by annotators, and we use metrics like Cohen's kappa coefficient to quantify the degree of agreement between annotations. When differences occur amongst annotators, we encourage cooperative attempts to settle disagreements by consensus-building and discussion. By working together, we can make sure that every instance has a single, consistent annotation, preserving the reliability and quality of our annotated dataset. Cohen's kappa (κ) and Fleiss kappa (κ_F) are statistical measures used to assess the level of agreement between two or more annotators beyond what would be expected by chance.

For Cohen's kappa, given N annotators and n items being annotated, it is calculated as:

$$\kappa = \frac{P(A) - P(E)}{1 - P(E)} \tag{8}$$

where:

- $P(A) = \frac{1}{N} \sum_{i=1}^{N} P(a_i)$ (the observed agreement)
- $P(E) = \frac{1}{N(N-1)} \sum_{i=1}^{N} \sum_{j=1, j\neq i}^{N} P(a_i)P(a_j)$ (the expected agreement)

$P(a_i)$ is the relative observed agreement among annotators for the ith category.

For Fleiss kappa, which extends Cohen's kappa to more than two annotators, it is calculated as:

$$\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e} \tag{9}$$

where:

- $\bar{P} = \frac{1}{Nn} \sum_{i=1}^{N} \sum_{k=1}^{K} n_{ik}(n_{ik} - 1)$ (the observed agreement)
- $\bar{P}_e = \frac{1}{n(n-1)} \sum_{k=1}^{K} \left(\sum_{i=1}^{N} n_{ik}^2 - n \right)$ (the expected agreement)

n_{ik} is the number of annotators who assigned the ith item to the kth category, N is the total number of annotators, n is the total number of items, and K is the total number of categories.

In our study, we evaluated the reliability of categorizing different types of fake news using Cohen's Kappa and Fleiss Kappa metrics. The Table 7 presents these metrics for four categories: Misinformation, Rumor, Clickbait, and Non-fake news. Cohen's Kappa measures the agreement between two raters, while Fleiss Kappa extends this to multiple raters. Higher values of Kappa indicate stronger agreement beyond chance. Across our dataset, we observed substantial agreement, with Cohen's Kappa ranging from 0.81 to 0.92 and Fleiss Kappa from 0.82 to 0.94. The overall Cohen's Kappa of 0.85 and Fleiss Kappa of 0.87 indicate robust agreement among evaluators in classifying fake news types. These metrics provide valuable insights into the consistency and reliability of classifications, crucial for understanding the nuances in detecting different forms of misinformation and real news.

Step (4) Annotation Quality Maintenance: We place a high priority on preserving annotation quality by using strict quality control processes. By keeping a close eye on the annotation process, we can quickly spot any irregularities or inconsistent labeling. Annotators receive timely feedback to rectify any concerns and guarantee adherence to the annotation criteria. Furthermore, the annotation process is supervised by a designated reviewer who confirms the accuracy and coherence of annotations. The annotated dataset is subjected to periodic assessments in order to maintain its quality. Moreover, we use refinement iterations, in which the annotation guideline is iteratively improved upon in response to comments from reviewers and annotators. The quality and dependability of the annotated dataset are eventually improved by this iterative approach, which guarantees that updates and revisions are performed to resolve any ambiguities or difficulties discovered throughout the annotation process.

Step (5) Annotation Validation and Bias Mitigation: Building upon our annotation process outlined in Step 3, we implemented rigorous validation procedures to ensure high-quality annotations and address potential biases in our dataset.

1. **Annotation Validation**
   Each sample in our dataset was independently evaluated by five annotators to minimize individual bias and increase reliability. This multi-annotator approach allowed us to establish a consensus-based labeling system and identify samples with high disagreement for further review.
2. **Bias Mitigation Strategies**
   To address potential biases in our annotation process:
   - **Annotator Diversity:** We ensured our team of annotators represented diverse backgrounds, ages, and political perspectives to reduce cultural and ideological biases in labeling.
   - **Blind Review Process:** Annotators were not informed about others' decisions to prevent anchoring bias and groupthink.
   - **Regular Calibration:** We conducted periodic calibration sessions where annotators collectively reviewed challenging examples to align on annotation standards.
   - **Bias Awareness Training:** Annotators received training on recognizing and mitigating personal biases when evaluating Bangla news content.
   - **Contextual Considerations:** We provided annotators with cultural and regional context for news stories to ensure proper interpretation of Bangla-specific references and idioms.
3. **Resolution of Disagreements**
   When significant disagreements occurred among annotators (defined as at least two divergent classifications), we implemented a structured resolution process:
   - All five annotators participated in a moderated discussion session.
   - Each annotator explained their reasoning without knowing others' initial judgments.
   - After discussion, annotators had the opportunity to revise their classifications.
   - If consensus could not be reached, senior annotators with expertise in the specific domain would make the final determination.

Table 8
Statistical overview of text–image pair data across different categories of fake news.

Category	Training	Testing	Validation
Entertainment	640	80	80
Sports	640	80	80
Technology	640	80	80
National	640	80	80
Lifestyle	640	80	80
Politics	640	80	80
Education	640	80	80
International	640	80	80
Crime	640	80	80
Finance	640	80	80
Business	640	80	80
Miscellaneous	640	80	80
Total	7680	960	960

Table 9
Distribution of text–image pair data within labels.

Label	Training	Testing	Validation
1	3840	480	480
0	3840	480	480
Total	7680	960	960

Table 10
Statistical overview of text–image pair data across different types of fake news.

Types	Training	Testing	Validation
Misinformation	1288	161	162
Rumor	1215	152	151
Clickbait	1337	167	167
Non-fake	3840	480	480
Total	7680	960	960

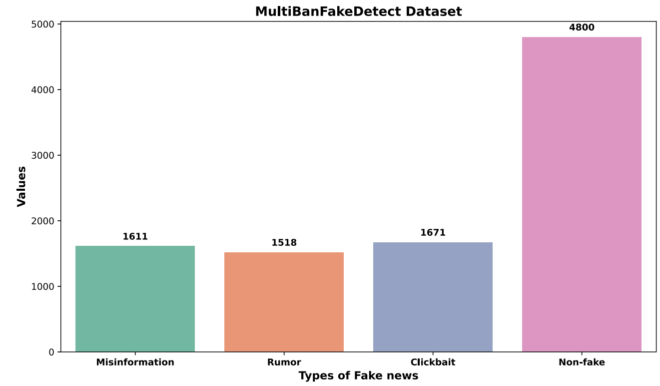

Fig. 4. The bar chart depicts an overview of text–image pair data across different types of fake news, illustrating key distributions in the dataset.

4.4. Dataset splitting

To ensure a balanced and representative dataset for multimodal fake news detection, we adopted a stratified dataset splitting approach with an **80% 10% 10% division** for training, validation, and testing. Given the presence of multiple categories—**clickbait, misinformation, rumor, and non-fake**—and the multiclass classification nature of our task, stratification was crucial in preserving equal label distribution across all subsets. To mitigate class imbalance and maintain a fair distribution, we applied **stratified sampling**, ensuring that each subset retained the same class proportions as the original dataset. Additionally, to enhance model robustness and minimize bias, we incorporated **k-fold cross-validation**. The training set was divided into k folds, and the model was iteratively trained on k-1 folds, while validating on the remaining fold. This process was repeated k times, allowing each instance to be used for both training and validation. Since our dataset supports both binary and multiclass classification, stratification was applied at both levels to prevent any class from being underrepresented. This structured approach, combined with k-fold cross-validation, ensures a balanced, diverse, and unbiased dataset, making it well-suited for training robust models for multimodal fake news detection (see Fig. 4 and Tables 8–10).

4.5. Comparative analysis with existing datasets

We compare our dataset's content, size, variety, and language coverage to those of other datasets that are currently available. In order to evaluate the scope and quality of our contribution, we carefully compare these features with other datasets that are made publically available. By performing this analysis, we hope to emphasize the special qualities and benefits of our dataset while also pointing out possible strengths and areas for development in other datasets. By placing our dataset within the larger context of fake news detection datasets, this thorough comparison enables us to highlight the importance and relevance of our dataset in furthering this field's study. Table 11 offers a thorough comparison study, contrasting our dataset's attributes with those of other datasets in the false news detection domain.

4.6. Addressing challenges in dataset creation

In our research, we encountered several challenges related to sourcing, understanding, and organizing fake data. These obstacles required us to develop innovative strategies to ensure the accuracy and comprehensiveness of our dataset. Below, we outline our approach to addressing these challenges in detail:

(a) Finding the fake data: We encountered challenges in sourcing fake data as fake pages and websites often get reported and shut down. To address this, we adopted a proactive approach by continuously monitoring online platforms and leveraging diverse sources to identify fake news instances.

(b) Understanding what is fake data and its varieties: We recognized the importance of comprehensively understanding different types of fake data and their characteristics. Through extensive research and analysis, we developed a nuanced understanding of various forms of misinformation, including clickbait, rumors, and fabricated content. This understanding informed our annotation guidelines and facilitated accurate identification and labeling of fake news instances within the dataset.

(c) Collecting data category-wise and topic-wise: To address the challenge of organizing data category-wise and topic-wise, we implemented systematic data collection strategies. We categorized topics based on relevance and significance, ensuring comprehensive coverage across different domains. By leveraging advanced search algorithms and keyword filters, we efficiently collected data tailored to specific categories and topics, optimizing the dataset for research purposes.

(d) Time consumption: Dataset creation can be time-consuming, especially when collecting and annotating large volumes of data. To mitigate this challenge, we employed efficient data collection and annotation workflows, leveraging automation and parallel processing techniques wherever possible. Additionally, we allocated sufficient resources and manpower to expedite the dataset creation process without compromising quality.

(e) Difficulty in finding fake news for certain topics: We encountered challenges in finding fake news for specific topics, especially those with limited availability or low prevalence of fake content. To address this, we expanded our search parameters and explored alternative sources to identify relevant fake news instances. Additionally, we engaged with domain experts and conducted extensive literature reviews to uncover hidden or less prominent instances of fake news within targeted topics.

Table 11
Comprehensive evaluation of the MultiBanFakeDetect dataset, Comparing its performance and characteristics against established datasets in the domain of fake news detection.

Dataset name	Ref	Text modality	Image modality	Fake news instances	Real news instances	Total instances	Dataset language	Year published
MultiBanFakeDetect	Proposed work	Yes	Yes	4800	4800	9600	Bangla	2024
Fakeddit	Kai Nakamura and Wang (2020)	Yes	Yes	414,088	268,908	682,996	English	2020
TwitterMediaEval2016	Boididou et al. (2016)	Yes	Yes	9596	6225	15 821	English	2016
PolitiFact	Shu, Sliva, Wang, Tang, and Liu (2017)	Yes	Yes	2140	1333	3473	English	2017
GossipCop	Shu, Mahudeswaran, Wang, Lee, and Liu (2020)	Yes	Yes	5323	16,817	22,140	English	2017
BanFakeNews	Hossain et al. (2020)	Yes	No	1299	48 678	49 977	Bangla	2020
Not mentioned	Hussain, Hasan, Rahman, Protim, and Hasan (2020b)	Yes	No	993	1548	2541	Bangla	2020
IFND	Sharma and Garg (2023)	Yes	Yes	19,059	37,809	56 868	English	2023

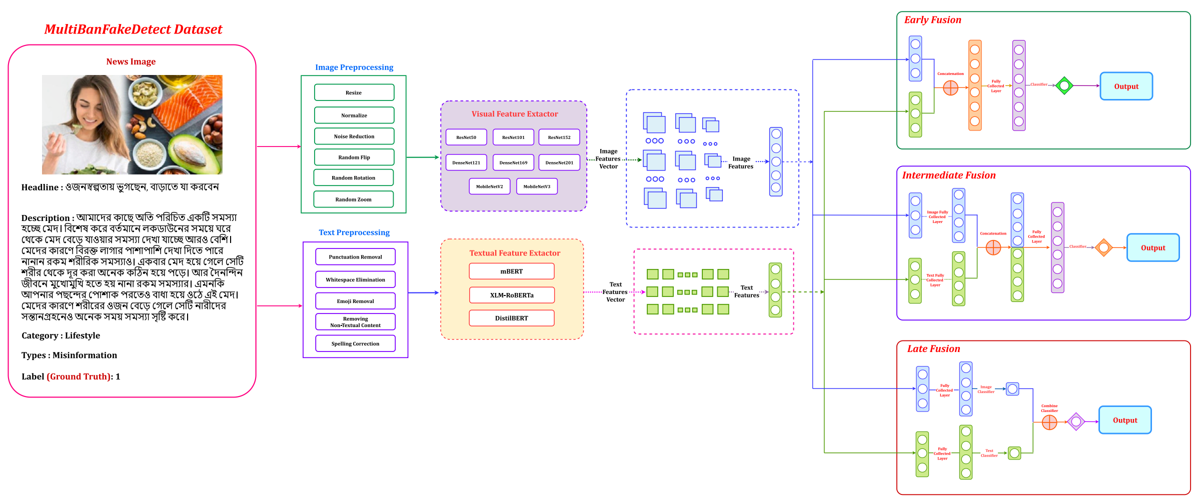

Fig. 5. The diagram depicts the MultiFusionFake framework, which incorporates three fusion strategies: Early Fusion, Intermediate Fusion, and Late Fusion.

(f) Separating data categorically: We recognized the importance of categorizing data accurately to facilitate precise analysis and modeling. To address this, we developed a systematic approach to categorize data into distinct categories such as clickbait, misinformation, and rumors. This involved designing robust annotation guidelines and implementing stringent quality assurance measures to ensure consistency and accuracy in data labeling.

4.7. Accessibility and utilization of the dataset

We prioritize accessibility and utilization of the dataset to facilitate widespread research in multimodal Bangla fake news detection. To achieve this, we make the dataset openly available to the research community, ensuring ease of access and dissemination. Detailed documentation and resources are provided to aid researchers in understanding the dataset structure, annotations, and potential applications.

5. Proposed methodology

We present a comprehensive methodology consisting of three distinct pipelines for Bangla fake news detection: the **TextFakeNet** framework for text-based detection and the **MultiFusionFake** framework for multimodal fake news detection that integrates both text and image modalities. Fig. 5 depicts the MultiFusionFake framework. The text-based approach (Unimodal) is expected to take approximately 30 to 45 min, focusing on text alone for detecting fake news. The image-based (Multimodal) approach, which applies various fusion techniques such as early, late, and intermediate fusion, is expected to take approximately 1 to 2 h. These approaches are designed to leverage the strengths of both textual and visual information for more robust fake news detection. The codes and supplementary coding files for this study are open to everyone publicly at: GitHub[1] platform.

5.1. Phase 1 for detecting bangla fake news (unimodal)

5.1.1. Text-based fake news detection: Binary classification (balanced dataset)

Step (1) Text Preprocessing:

Text preprocessing is an essential first step in preparing Bangla text for the training of our fake news detection model. The primary goal of this step is to ensure that the text is clean, standardized, and ready for effective model training. To achieve this, we carried out several preprocessing tasks to improve the quality of the text:

- **Emoji Removal:** Emojis, which often express emotions or concepts without contributing to the text's meaning, are removed to avoid unnecessary noise in the data.
- **Whitespace Handling:** We meticulously cleaned all forms of whitespace, including spaces, tabs, and line breaks, to ensure consistency in the text formatting.
- **Punctuation Removal:** All punctuation marks such as commas, periods, exclamation marks, and question marks are stripped away, as they offer little to no value in understanding the semantics and can interfere with model training.
- **Special Symbols Removal:** Any special symbols, including currency symbols, mathematical signs, and diacritics, are eliminated since they could hinder the model's comprehension of the text.

[1] MultiBanFakeDetect-An-Extensive-Benchmark-Dataset-for-Multimodal-Bangla-Fake-News-Detection.

Table 12
Hyperparameter settings for text-based models in Bangla fake news detection.

Approach	Model	Batch size	Epoch	Learning rate	Optimizer
Text based	TextFakeNet	16	15	1e−3	AdamW
	XLM-RoBERTa	16	15	1e−3	AdamW
	DistilBERT	16	15	1e−3	AdamW

These preprocessing steps collectively ensure that the data is well-structured and consistent, which is vital for achieving optimal performance in the binary classification task.

Step (2) Bangla Fake News Detection Model Development:

In this phase, we develop a robust Bangla Fake News Detection model utilizing cutting-edge, pre-trained language models. Given the task of binary classification (Fake vs. Non-Fake News), we harness powerful multilingual models such as **multilingual BERT (mBERT)**, **XLM-RoBERTa**, and **DistilBERT**. These models have been pre-trained on large-scale, multilingual corpora, which allows them to capture complex linguistic patterns and nuances across various languages, including Bangla.

By fine-tuning these pre-trained models on our balanced Bangla fake news dataset, we aim to effectively distinguish between fake and non-fake news articles. The fine-tuning process allows the models to leverage their advanced understanding of semantic and contextual elements in Bangla texts, making them highly proficient in identifying fake news articles.

Step (3) Hyperparameter Tuning:

Hyperparameter tuning is a pivotal step in enhancing the performance of our fake news detection models. It involves fine-tuning the models' parameters to achieve optimal results in terms of accuracy, precision, recall, and F1-score. During this process, we experiment with and adjust hyperparameters such as the learning rate, batch size, dropout rate, and regularization strength.

By conducting rigorous trials and validation, we iteratively optimize these hyperparameters to ensure the best possible performance on the balanced dataset. The results of this tuning process are meticulously documented and summarized in Table 12, providing valuable insights into how different settings impact model performance. This comprehensive analysis helps identify the most effective configuration for each model, leading to peak performance in binary classification tasks such as distinguishing between fake and non-fake news.

Step (4) Evaluation of Bangla Fake News Detection:

After model development and hyperparameter tuning, we proceed with a thorough evaluation of our models' performance. To assess the effectiveness of our Bangla Fake News Detection models, we use a range of performance metrics, including **accuracy**, **precision**, **recall**, and **F1-score**. These metrics provide a detailed view of the model's ability to correctly classify both fake and non-fake news articles, ensuring that genuine news is not misclassified as fake, and vice versa.

The evaluation results are carefully analyzed and presented in Table 16, which offers a comprehensive breakdown of the performance for each model variant. This analysis allows us to evaluate the strengths and weaknesses of our models, offering clear guidance for future optimization and improvements. The insights gained from this evaluation are essential in refining our models to achieve the highest levels of accuracy in fake news detection for Bangla texts.

5.1.2. Text-based fake news detection: Multiclass classification (imbalanced dataset)

Step (1) Text Preprocessing:

For text preprocessing, we followed the same steps outlined in Section 5.1.1, ensuring consistency and applying the same rigorous approach used in the binary classification task, tailored to the multiclass setting.

Step (2) Bangla Fake News Detection Model Development:

In this phase, we develop a robust Bangla Fake News Detection model utilizing cutting-edge, pre-trained language models. Given the task of multiclass classification (clickbait, misinformation, rumor, non-fake), we harness powerful multilingual models such as **multilingual BERT (mBERT)**, **XLM-RoBERTa**, and **DistilBERT**. These models have been pre-trained on large-scale, multilingual corpora, which allows them to capture complex linguistic patterns and nuances across various languages, including Bangla.

By fine-tuning these pre-trained models on our imbalanced Bangla fake news dataset, we aim to effectively classify news articles into one of the four categories: clickbait, misinformation, rumor, or non-fake. Fine-tuning allows the models to learn intricate semantic and contextual relationships in Bangla texts, improving their ability to identify fake content with high precision.

Step (3) Cost-Sensitive Approach for Handling Imbalanced Dataset:

Given the class imbalance in our dataset, where certain categories such as clickbait or misinformation are significantly more frequent than others like rumor or non-fake, it is crucial to employ a **cost-sensitive learning approach**. In this approach, we assign higher misclassification costs to the minority classes, ensuring that the model pays more attention to these underrepresented categories. This helps in improving the overall performance across all classes, especially those that are typically underrepresented in imbalanced datasets. A typical cost-sensitive loss function can be defined as:

$$\mathcal{L}_{cs}(\hat{y}, y) = \sum_{i=1}^{C} c_i \cdot \mathbb{I}(y = i) \cdot \log(\hat{p}_i) \tag{10}$$

where: - C is the number of classes (4 in our case: clickbait, misinformation, rumor, non-fake), - $\hat{y}$ is the predicted label, - y is the true label, - $\hat{p}_i$ is the predicted probability for class i, - c_i is the misclassification cost for class i, - $\mathbb{I}(y = i)$ is the indicator function that is 1 if $y = i$, and 0 otherwise.

The misclassification cost c_i is typically defined as the inverse of the class frequency or a pre-defined weight to reflect the severity of misclassifying a minority class. In the case of imbalanced datasets, we use higher values for minority classes (such as rumor and non-fake) to ensure they are adequately represented during model training.

Step (4) Evaluation of Bangla Fake News Detection:

After model development and the implementation of the cost-sensitive approach, we evaluate the performance of our Bangla Fake News Detection models. We employed a comprehensive set of performance metrics including accuracy, precision, recall, and F1-score to assess the effectiveness of our models in distinguishing between the four categories: clickbait, misinformation, rumor, and non-fake. These metrics provide insights into the model's ability to correctly identify each class, particularly focusing on recall to ensure that minority classes (like rumor and non-fake) are correctly classified. The evaluation results were meticulously analyzed and are presented in Table 17, offering a detailed breakdown of the performance metrics for each model variant. This thorough analysis enables us to gain a deeper understanding of the strengths and weaknesses of our fake news detection models and informs further optimization and refinement.

5.2. Phase 2 for exploring fusion strategies in bangla fake news detection (multimodal)

5.2.1. Multimodal fake news classification (binary: Fake vs. Non-fake)

Step (1) Feature Extraction: Feature Extraction forms the cornerstone of the Multimodal Fake News Classification pipeline. For textual features, we harnessed the capabilities of advanced pre-trained language models such as mBERT, XLM-RoBERTa, and DistilBERT. These models have been pre-trained on extensive text corpora, enabling them to effectively capture intricate linguistic patterns and semantic relationships within text. Simultaneously, we extract image features using a diverse range of state-of-the-art convolutional CNN models, including ResNet 50, ResNet 101, ResNet 152, DenseNet 121, DenseNet 169, DenseNet 201, MobileNet V2, and MobileNet V3. These CNN architectures are well-regarded for their ability to extract high-level image representations, capturing complex visual features and patterns present in images. By leveraging both textual and image features, our approach aims to capture complementary information from different modalities. This comprehensive integration enhances our ability to understand multimodal data and improves the overall performance of our Multimodal Fake News Classification system.

Step (2) Fusion Techniques: We applied three different fusion techniques to seamlessly integrate textual and visual information in our Multimodal Fake News Classification pipeline. Before exploring these techniques, we lay the groundwork for extracting rich features from both textual and visual modalities. This ensures that the essential information in each modality is captured. These fusion approaches enable our model to leverage the complementary nature of text and image data, thereby enhancing its ability to classify fake and non-fake news in a multimodal context.

(a) ***Early Fusion for Multimodal Fake News Classification:*** In the early fusion approach (Barnum, Talukder, & Yue, 2020), text and image features were combined at the input level before being processed by the subsequent layers of the model. This integration occurs seamlessly, allowing the model to jointly learn from both modalities from the outset. By merging text and image features early in the pipeline, the model can capture the interactions and correlations between textual and visual information, potentially leading to more robust representations of the multimodal data.

$$z_{\text{early}} = f_{\text{early}}(\phi_{\text{text}}(f_{\text{text}}(x_{\text{text}})), \phi_{\text{image}}(f_{\text{image}}(x_{\text{image}}))) \tag{11}$$

(b) ***Late Fusion for Multimodal Fake News Classification:*** Unlike early fusion, the late fusion (Pandeya & Lee, 2021) technique involves processing text and image features separately through independent pathways before merging them at a later stage of the model. This approach allows each modality to undergo individualized processing, capturing unique characteristics and patterns specific to text and images. The fused representations are then combined in a subsequent layer, enabling the model to integrate the information from both modalities while preserving their distinct features and nuances.

$$z_{\text{late}} = f_{\text{combine}}(\phi_{\text{text}}(f_{\text{text}}(x_{\text{text}})), \phi_{\text{image}}(f_{\text{image}}(x_{\text{image}}))) \tag{12}$$

(c) ***Intermediate Fusion for Multimodal Fake News Classification:*** Intermediate fusion (Hu, Jia, & Rostami, 2024) strikes a balance between early and late fusion by combining text and image features in an intermediate layer of the model. This allows for a more flexible integration of multimodal information, as the fusion occurs after the initial processing of each modality but before the final output. Intermediate fusion enables the model to leverage both early and late fusion benefits, capturing both low-level and high-level interactions between text and image features while maintaining a degree of modality-specific processing.

$$\begin{aligned} z_{\text{intermediate}} = f_{\text{intermediate}} \Big(& \phi_{\text{early}}(f_{\text{early}}(\phi_{\text{text}}(f_{\text{text}}(x_{\text{text}})), \\ & \phi_{\text{image}}(f_{\text{image}}(x_{\text{image}})))) \Big) \end{aligned} \tag{13}$$

Step (3) Hyperparameter Tuning: In this phase, we focused on optimizing the performance of our multimodal fake news classification model by fine-tuning key hyperparameters. The aim was to strike a balance between training efficiency, model stability, and generalization ability. The key hyperparameters tested included batch size, epochs, learning rate, and optimizer. Below are the specific details of the tuning process:

- **Batch Size (8):** A batch size of 8 was chosen for its ability to provide a balance between training speed and model stability. Larger batch sizes, such as 16 or 32, led to higher memory usage and sometimes introduced instability in the training process. The batch size of 8 allowed for stable updates while maintaining efficient use of resources.
- **Epochs (30–45):** We experimented with a range of epochs between 30 and 45. Our results indicated that the model performed optimally within the 35–40 epoch range. During this range, validation accuracy plateaued, and the model exhibited no signs of overfitting, ensuring a good balance between training time and model generalization.
- **Learning Rate (1e−4, 1e−3):** We experimented with two learning rates: 1e−4 and 1e−3, testing each across different models to evaluate their impact on training stability and performance. With a learning rate of 1e−3, the models showed faster convergence, which was beneficial for reducing training time. However, we also observed that this learning rate occasionally led to training instability, especially during the later stages of optimization, where the model struggled to fine-tune weights without significant fluctuations. On the other hand, a learning rate of 1e−4 provided more stability during training, allowing the models to reach a good level of generalization. Despite the more stable training, the convergence was slower compared to the 1e−3 rate. After testing both learning rates extensively across various models, we found that while 1e−4 was more stable, 1e−3 consistently provided better overall performance in terms of both training speed and validation accuracy. As a result, we selected the learning rate of 1e−3 for our final model, as it offered the best trade-off between convergence speed and model performance.
- **Optimizer (AdamW):** The AdamW optimizer was selected for its superior performance in handling large datasets and multimodal inputs. Unlike the standard Adam optimizer, AdamW incorporates weight decay directly into the optimization process, which helped improve model stability and reduce overfitting. This made it the optimal choice for our multimodal classification task.

The following tables summarize the outcomes of our extensive hyperparameter tuning process across different fusion strategies. Table 13 presents the tuning results for early fusion strategies, focusing on the effects of various learning rates, batch sizes, and regularization techniques on the model's ability to combine features from both modalities at the input stage. Table 15 shows the outcomes of late fusion approaches, where the model integrates outputs from independent modality-specific models, highlighting how fine-tuning hyperparameters like learning rates and regularization strength impacted the performance. Finally, Table 14 details the intermediate fusion strategy, where textual and visual data are combined at hidden layers, and illustrates how fusion layer configurations interact with the selected hyperparameters.

Step (4) Evaluation of Multimodal Fake News Classification: In this phase, we analyze different performance measures specifically designed for multimodal classification to gauge how well our Multimodal Fake News Classification system performs. We measured metrics such as accuracy, precision, recall, and F1-score to assess how well our algorithms work when identifying fake and non-fake news from both textual and visual sources. The strengths and shortcomings of our Multimodal Fake News Classification system are shown through a thorough analysis of the evaluation findings. Tables 18, 20, and 22 provide a thorough analysis of the evaluation results and breakdown of the performance metrics for each fusion procedure.

5.2.2. Multimodal fake news classification (Clickbait, misinformation, rumor, non-fake) with cost-sensitive learning

Step (1) Feature Extraction: Feature Extraction forms the cornerstone of the Multimodal Fake News Classification pipeline. For textual

Table 13
Hyperparameter settings for early fusion in multimodal models for Bangla fake news detection.

Approach	Model	Batch size	Epoch	Learning rate	Optimizer
Early fusion	ResNet50+mBERT	8	40	1e−4	AdamW
	ResNet101+mBERT	8	35	1e−4	AdamW
	ResNet152+mBERT	8	45	1e−4	AdamW
	DenseNet121+mBERT	8	30	1e−3	AdamW
	MultiFusionFake	8	30	1e−3	AdamW
	DenseNet201+mBERT	8	35	1e−3	AdamW
	MobileNetV2+mBERT	8	40	1e−4	AdamW
	MobileNetV3++mBERT	8	40	1e−4	AdamW
	ResNet50+XLM-RoBERTa	8	45	1e−4	AdamW
	ResNet101+XLM-RoBERTa	8	45	1e−4	AdamW
	ResNet152+XLM-RoBERTa	8	40	1e−4	AdamW
	DenseNet121+XLM-RoBERTa	8	35	1e−3	AdamW
	DenseNet169+XLM-RoBERTa	8	35	1e−3	AdamW
	DenseNet201+XLM-RoBERTa	8	30	1e−3	AdamW
	MobileNetV2+XLM-RoBERTa	8	30	1e−4	AdamW
	MobileNetV3+XLM-RoBERTa	8	30	1e−4	AdamW
	ResNet 50+DistilBERT	8	40	1e−4	AdamW
	ResNet 101+DistilBERT	8	35	1e−3	AdamW
	ResNet 152+DistilBERT	8	40	1e−4	AdamW
	DenseNet 121+DistilBERT	8	40	1e−4	AdamW
	DenseNet 169+DistilBERT	8	35	1e−3	AdamW
	DenseNet 201+DistilBERT	8	40	1e−4	AdamW
	MobileNet V2+DistilBERT	8	40	1e−4	AdamW
	MobileNet V3+DistilBERT	8	35	1e−3	AdamW

Table 14
Hyperparameter settings for intermediate fusion in multimodal models for Bangla fake news detection.

Approach	Model	Batch size	Epoch	Learning rate	Optimizer
Intermediate fusion	ResNet50+mBERT	8	40	1e−4	AdamW
	ResNet101+mBERT	8	35	1e−4	AdamW
	ResNet152+mBERT	8	45	1e−4	AdamW
	DenseNet121+mBERT	8	30	1e−3	AdamW
	DenseNet169+mBERT	8	30	1e−3	AdamW
	DenseNet201+mBERT	8	35	1e−3	AdamW
	MobileNetV2+mBERT	8	40	1e−4	AdamW
	MobileNetV3++mBERT	8	40	1e−4	AdamW
	ResNet50+XLM-RoBERTa	8	45	1e−4	AdamW
	ResNet101+XLM-RoBERTa	8	45	1e−4	AdamW
	ResNet152+XLM-RoBERTa	8	40	1e−4	AdamW
	DenseNet121+XLM-RoBERTa	8	35	1e−3	AdamW
	DenseNet169+XLM-RoBERTa	8	35	1e−3	AdamW
	DenseNet201+XLM-RoBERTa	8	35	1e−3	AdamW
	MobileNetV2+XLM-RoBERTa	8	40	1e−4	AdamW
	MobileNetV3+XLM-RoBERTa	8	35	1e−4	AdamW
	ResNet 50+DistilBERT	8	40	1e−4	AdamW
	ResNet 101+DistilBERT	8	35	1e−3	AdamW
	ResNet 152+DistilBERT	8	40	1e−4	AdamW
	DenseNet 121+DistilBERT	8	40	1e−4	AdamW
	DenseNet 169+DistilBERT	8	35	1e−3	AdamW
	DenseNet 201+DistilBERT	8	40	1e−4	AdamW
	MobileNet V2+DistilBERT	8	40	1e−4	AdamW
	MobileNet V3+DistilBERT	8	35	1e−3	AdamW

features, we harness the capabilities of advanced pre-trained language models such as mBERT, XLM-RoBERTa, and DistilBERT. These models have been pre-trained on extensive text corpora, enabling them to effectively capture intricate linguistic patterns and semantic relationships within text. Simultaneously, we extract image features using a diverse range of state-of-the-art convolutional CNN models, including ResNet 50, ResNet 101, ResNet 152, DenseNet 121, DenseNet 169, DenseNet 201, MobileNet V2, and MobileNet V3. These CNN architectures are well-regarded for their ability to extract high-level image representations, capturing complex visual features and patterns present in images. By leveraging both textual and image features, our approach aims to capture complementary information from different modalities. This comprehensive integration enhances our ability to understand multimodal data and improves the overall performance of our Multimodal Fake News Classification system.

Step (2) Cost-Sensitive Fusion Techniques: We applied three different cost-sensitive fusion techniques to seamlessly integrate textual and visual information in our Multimodal Fake News Classification pipeline. Given the class imbalance issue in our dataset, we incorporate cost-sensitive learning by introducing class-dependent weighting factors into the fusion process. This ensures that the model does not become biased toward majority classes and effectively learns from underrepresented categories.

(a) ***Early Fusion with Cost Sensitivity:*** In the early fusion approach (Barnum et al., 2020), text and image features were combined at the input level before being processed by subsequent layers of the model. We introduce class-dependent weighting matrices, denoted as W_c, which adjust feature representations according to the relative importance of each class in the dataset.

$$z_{\text{early}} = W_c \cdot f_{\text{early}}(\phi_{\text{text}}(f_{\text{text}}(x_{\text{text}})), \phi_{\text{image}}(f_{\text{image}}(x_{\text{image}}))) \quad (14)$$

where:

- W_c is the cost-sensitive weight matrix that adjusts feature importance based on class distribution.

Table 15
Hyperparameter settings for late fusion in multimodal models for Bangla fake news detection.

Approach	Model	Batch size	Epoch	Learning rate	Optimizer
Late fusion	ResNet50+mBERT	8	35	1e−4	AdamW
	ResNet101+mBERT	8	35	1e−4	AdamW
	ResNet152+mBERT	8	40	1e−4	AdamW
	DenseNet121+mBERT	8	30	1e−3	AdamW
	DenseNet169+mBERT	8	30	1e−3	AdamW
	DenseNet201+mBERT	8	35	1e−3	AdamW
	MobileNetV2+mBERT	8	40	1e−4	AdamW
	MobileNetV3++mBERT	8	40	1e−4	AdamW
	ResNet50+XLM-RoBERTa	8	40	1e−4	AdamW
	ResNet101+XLM-RoBERTa	8	40	1e−4	AdamW
	ResNet152+XLM-RoBERTa	8	40	1e−4	AdamW
	DenseNet121+XLM-RoBERTa	8	35	1e−3	AdamW
	DenseNet169+XLM-RoBERTa	8	35	1e−3	AdamW
	DenseNet201+XLM-RoBERTa	8	30	1e−3	AdamW
	MobileNetV2+XLM-RoBERTa	8	30	1e−4	AdamW
	MobileNetV3+XLM-RoBERTa	8	30	1e−4	AdamW
	ResNet 50+DistilBERT	8	40	1e−4	AdamW
	ResNet 101+DistilBERT	8	35	1e−3	AdamW
	ResNet 152+DistilBERT	8	40	1e−4	AdamW
	DenseNet 121+DistilBERT	8	40	1e−4	AdamW
	DenseNet 169+DistilBERT	8	35	1e−3	AdamW
	DenseNet 201+DistilBERT	8	40	1e−4	AdamW
	MobileNet V2+DistilBERT	8	40	1e−4	AdamW
	MobileNet V3+DistilBERT	8	35	1e−3	AdamW

- x_{text} and x_{image} are the input textual and visual data, respectively.
- $f_{\text{text}}(\cdot)$ and $f_{\text{image}}(\cdot)$ are the feature extraction functions for text and images.
- $\phi_{\text{text}}(\cdot)$ and $\phi_{\text{image}}(\cdot)$ are non-linear activation functions applied to the extracted features.
- $f_{\text{early}}(\cdot,\cdot)$ is the early fusion function.
- z_{early} represents the fused features after cost-sensitive early fusion.

(b) ***Late Fusion with Cost Sensitivity:*** In the late fusion approach (Pandeya & Lee, 2021), text and image features were processed separately and merged at a later stage. We apply cost-sensitive weighting at the decision-level fusion by introducing a penalty matrix λ_c that scales each class prediction accordingly.

$$z_{\text{late}} = \lambda_c \cdot f_{\text{combine}}(\phi_{\text{text}}(f_{\text{text}}(x_{\text{text}})), \phi_{\text{image}}(f_{\text{image}}(x_{\text{image}}))) \quad (15)$$

where:

- λ_c is the penalty factor applied to balance class contributions.
- Other variables are as defined in the early fusion equation.

(c) ***Intermediate Fusion with Cost Sensitivity:*** Intermediate fusion (Hu et al., 2024) integrates multimodal features in an intermediate layer of the model, striking a balance between early and late fusion. We introduce class-dependent scaling factors γ_c to ensure balanced learning across different categories.

$$z_{\text{intermediate}} = \gamma_c \cdot f_{\text{intermediate}}\Big(\phi_{\text{early}}\big(f_{\text{early}}(\phi_{\text{text}}(f_{\text{text}}(x_{\text{text}}))), \phi_{\text{image}}(f_{\text{image}}(x_{\text{image}}))\big)\Big) \quad (16)$$

where:

- γ_c is the cost-sensitive weighting factor for class-dependent intermediate fusion.
- Other variables remain consistent with previous equations.

Step (3) Hyperparameter Tuning with Cost Sensitivity: We followed the same approach as described in Step 3 under Section 5.2.1, fine-tuning hyperparameters such as class balancing factors, learning rate, batch size, regularization strength, and fusion layer configurations. The optimized hyperparameters for each fusion technique, incorporating cost-sensitive factors to address class imbalance, are presented in Tables 13, 15, and 14.

Step (4) Evaluation of Multimodal Fake News Classification: We evaluate our model using accuracy, precision, recall, and F1-score metrics to account for class imbalance. Cost-sensitive loss functions such as weighted cross-entropy and focal loss are used to improve model performance. Tables 19, 21, and 23 provide a comparative analysis of different fusion strategies under cost-sensitive settings.

6. Experiments and result analysis

6.1. Experimental setup

The experiment was conducted in four different environments, two using Jupyter Notebook, one using Kaggle, and one using Google Colaboratory. One Jupyter Notebook setup employed an NVIDIA GeForce RTX 3050 GPU (computing capability 8.6), an Intel Core i5 9400f CPU, and 16 GB of RAM. The other Jupyter Notebook environment was equipped with a more advanced NVIDIA GeForce RTX 4060 Ti 16 GB GPU, AMD Ryzen 7 5700X processor, and 32 GB of RAM. Both Jupyter Notebook environments used Python version 3.8.18 with PyTorch version 2.0.1. On the other hand, Kaggle provide access to NVIDIA Tesla P100 GPUs (compute capability 6.0), Intel Xeon CPUs, and 12.72 GB of RAM. The Kaggle setup utilized Python 3.10.13 with PyTorch version 2.1.2. Finally, the Google Colaboratory setup utilized a Tesla T4 GPU (15 GB), 12.5 GB of RAM, and Python 3.10.12 with PyTorch 2.3.1.

6.2. Hyper-parameter settings

Tables 12, 13, 14, and 15 present the hyperparameter settings for text-based and multimodal models in Bangla Fake News Detection for both binary and multiclass classification. For text-based models (mBERT, XLM-RoBERTa, DistilBERT), the batch size is 16, epochs are set to 15, the learning rate is 1e−3, and the optimizer is AdamW. Early, intermediate, and late fusion multimodal models use a batch size of 8, with epochs ranging from 30 to 45 for early and intermediate fusion, and 30 to 40 for late fusion. The learning rate is either 1e−4 or 1e−3 across all multimodal models, with AdamW consistently used as the optimizer.

Table 16
Performance metrics for text-based models in Bangla fake news binary classification (5-Fold cross-validation, Mean ± Std).

Approach	Model	Accuracy	Precision	Recall	F1-Score
Text based	**TextFakeNet**	**0.7313 ± 0.012**	**0.7361 ± 0.014**	**0.7284 ± 0.011**	**0.7226 ± 0.013**
	XLM-RoBERTa	0.7054 ± 0.015	0.7143 ± 0.017	0.7092 ± 0.012	0.7118 ± 0.014
	DistilBERT	0.7146 ± 0.010	0.7152 ± 0.013	0.7146 ± 0.011	0.7144 ± 0.012

Table 17
Performance metrics for text-based models in Bangla fake news multiclass classification (5-Fold cross-validation, Mean ± Std).

Approach	Model	Accuracy	Precision	Recall	F1-Score
Text based	mBERT	0.6300 ± 0.018	0.6100 ± 0.017	0.6200 ± 0.016	0.6150 ± 0.015
	XLM-RoBERTa	0.6400 ± 0.020	0.6200 ± 0.019	0.6300 ± 0.018	0.6250 ± 0.017
	DistilBERT	0.6000 ± 0.015	0.6000 ± 0.014	0.6100 ± 0.013	0.6050 ± 0.012

Table 18
Performance metrics for multimodal-based models with early fusion in Bangla fake news binary classification (5-fold cross-validation, Mean ± Std).

Approach	Model	Accuracy	Precision	Recall	F1-Score
	ResNet50+mBERT	0.7813 ± 0.014	0.8750 ± 0.022	0.7368 ± 0.018	0.8000 ± 0.020
	ResNet101+mBERT	0.7604 ± 0.012	0.8333 ± 0.019	0.7273 ± 0.020	0.7767 ± 0.021
	ResNet152+mBERT	0.7083 ± 0.017	0.7292 ± 0.020	0.7174 ± 0.021	0.7143 ± 0.018
	DenseNet121+mBERT	0.7031 ± 0.011	0.7083 ± 0.017	0.7010 ± 0.016	0.7047 ± 0.014
	MultiFusionFake	**0.7969 ± 0.013**	**0.7879 ± 0.014**	**0.8065 ± 0.012**	**0.7937 ± 0.015**
	DenseNet201+mBERT	0.7396 ± 0.012	0.7500 ± 0.013	0.7347 ± 0.015	0.7423 ± 0.014
	MobileNetV2+mBERT	0.7448 ± 0.015	0.7604 ± 0.016	0.7374 ± 0.014	0.7487 ± 0.015
	MobileNetV3+mBERT	0.7552 ± 0.016	0.7604 ± 0.014	0.7526 ± 0.016	0.7565 ± 0.015
	ResNet50+XLM-RoBERTa	0.7708 ± 0.011	0.7826 ± 0.013	0.7600 ± 0.012	0.7755 ± 0.013
	ResNet101+XLM-RoBERTa	0.7865 ± 0.012	0.8229 ± 0.015	0.7670 ± 0.013	0.7940 ± 0.014
	ResNet152+XLM-RoBERTa	0.7500 ± 0.014	0.7708 ± 0.017	0.7400 ± 0.015	0.7551 ± 0.016
Early fusion	DenseNet121+XLM-RoBERTa	0.7551 ± 0.012	0.7600 ± 0.011	0.7600 ± 0.012	0.7600 ± 0.013
	DenseNet169+XLM-RoBERTa	0.7885 ± 0.011	0.7959 ± 0.013	0.7911 ± 0.012	0.7935 ± 0.013
	DenseNet201+XLM-RoBERTa	0.7463 ± 0.012	0.7125 ± 0.015	0.7685 ± 0.014	0.7395 ± 0.013
	MobileNetV2+XLM-RoBERTa	0.7368 ± 0.011	0.7042 ± 0.014	0.7578 ± 0.013	0.7300 ± 0.012
	MobileNetV3+XLM-RoBERTa	0.7332 ± 0.013	0.6962 ± 0.015	0.7638 ± 0.014	0.7284 ± 0.014
	ResNet50+DistilBERT	0.7104 ± 0.015	0.7752 ± 0.017	0.7665 ± 0.016	0.7682 ± 0.015
	ResNet101+DistilBERT	0.7716 ± 0.011	0.7159 ± 0.014	0.7386 ± 0.013	0.7193 ± 0.012
	ResNet152+DistilBERT	0.7790 ± 0.013	0.7598 ± 0.012	0.7364 ± 0.013	0.7151 ± 0.015
	DenseNet121+DistilBERT	0.7348 ± 0.014	0.7360 ± 0.015	0.7683 ± 0.014	0.7609 ± 0.013
	DenseNet169+DistilBERT	0.7809 ± 0.013	0.7477 ± 0.015	0.7196 ± 0.014	0.7670 ± 0.012
	DenseNet201+DistilBERT	0.7708 ± 0.012	0.7548 ± 0.014	0.7716 ± 0.013	0.7495 ± 0.015
	MobileNetV2+DistilBERT	0.7518 ± 0.012	0.7442 ± 0.014	0.7120 ± 0.013	0.7186 ± 0.014
	MobileNetV3+DistilBERT	0.7125 ± 0.013	0.7608 ± 0.014	0.7351 ± 0.013	0.7506 ± 0.012

6.3. Quantitative analysis

6.3.1. Fake news binary classification

Table 16 compares mBERT and XLM-RoBERTa for Bangla fake news detection. mBERT outperformed XLM-RoBERTa with an accuracy of 73.13% vs. 70.54%, and a higher F1 Score (72.26% vs. 71.18%), showing slightly better overall effectiveness.

Table 18 presents multimodal early fusion results. **MultiFusion-Fake (DenseNet 169 + mBERT)** achieved the highest accuracy (79.69%) and F1 Score (79.37%), excelling in both precision and recall. ResNet 101 + XLM-RoBERTa followed closely. DenseNet 169 + XLM-RoBERTa also performed well. The weakest model was DenseNet 121 + mBERT, highlighting areas for improvement.

Table 20 compares intermediate fusion models for Bangla fake news detection. ResNet 101 + mBERT achieved the highest accuracy (79.27%) and F1 Score (80.00%), excelling in both precision (82.92%) and recall (77.28%). ResNet 50 + XLM-RoBERTa followed closely, while MobileNet V3 + XLM-RoBERTa had the lowest performance (74.69% accuracy, 74.77% F1 Score), indicating room for improvement.

Table 22 presents late fusion results. ResNet 101 + XLM-RoBERTa performed best, achieving the highest accuracy (78.44%) and F1 Score (78.68%). DenseNet 169 + mBERT followed closely. The weakest model, MobileNet V3 + XLM-RoBERTa, had an F1 Score of 74.45%, suggesting the need for better precision–recall balance.

In conclusion, our experimental results demonstrated notable accuracy rates across various approaches. **MultiFusionFake**, an early fusion approach using DenseNet 169 and mBERT, achieved the highest accuracy of 79.69%. Late Fusion with ResNet 101 and XLM-RoBERTa reached 78.44%, whereas Intermediate Fusion with ResNet 101 and mBERT reached 79.27%. In the text-based approaches, mBERT achieved 73.13% accuracy. These findings highlight the substantial improvement that multimodal approaches offer over unimodal methods, with the best multimodal model achieving a 6.56% higher accuracy compared to the text-based mBERT model. Ablation testing results, highlighting the performance comparison between the text-only (TextFakeNet) and the text+image fusion model (MultiFusionFake), are illustrated in Fig. 6.

6.3.2. Fake news multiclass classification

Table 17 presents the multiclass Bangla fake news detection results across different fusion approaches. In the text-based models, XLM-RoBERTa achieved the highest accuracy (64.00%) and F1 Score (63.80%), followed by mBERT with an accuracy of 63.00% and DistilBERT with 60.00%.

For early fusion Table 19, DenseNet 169 + mBERT performed best, achieving an accuracy of 73.80% and an F1 Score of 71.20%. DenseNet 121 + XLM-RoBERTa followed closely with an accuracy of 73.40% and an F1 Score of 72.60%. These results highlight the advantage of integrating textual and visual features early. Intermediate fusion models Table 21 further improved performance. MobileNet V3 + DistilBERT achieved an accuracy of 71.30% and an F1 Score of 71.70%, while MobileNet V2 + DistilBERT followed closely with an accuracy of 70.70% and an F1 Score of 71.10%. These findings indicate the benefit of deeper feature interactions. For late fusion Table 23, DenseNet 169 + XLM-RoBERTa achieved the highest accuracy (73.00%) and F1 Score

Table 19
Performance metrics for multimodal-based models with early fusion in Bangla fake news multiclass classification (5-fold cross-validation, Mean ± Std).

Approach	Model	Accuracy	Precision	Recall	F1-Score
Early fusion	ResNet50+mBERT	0.6520 ± 0.012	0.6090 ± 0.015	0.6520 ± 0.014	0.6210 ± 0.013
	ResNet101+mBERT	0.6400 ± 0.013	0.6000 ± 0.014	0.6400 ± 0.012	0.6120 ± 0.011
	ResNet152+mBERT	0.7340 ± 0.014	0.7210 ± 0.012	0.7320 ± 0.013	0.7260 ± 0.011
	DenseNet121+mBERT	0.7350 ± 0.013	0.7220 ± 0.014	0.7330 ± 0.015	0.7270 ± 0.013
	DenseNet169+mBERT	0.7380 ± 0.011	0.7000 ± 0.013	0.7380 ± 0.014	0.7120 ± 0.012
	DenseNet201+mBERT	0.6360 ± 0.012	0.5970 ± 0.014	0.6360 ± 0.015	0.6090 ± 0.014
	MobileNetV2+mBERT	0.6600 ± 0.013	0.6150 ± 0.015	0.6600 ± 0.014	0.6270 ± 0.012
	MobileNetV3+mBERT	0.6440 ± 0.011	0.6030 ± 0.013	0.6440 ± 0.015	0.6150 ± 0.012
	ResNet50+XLM-RoBERTa	0.6480 ± 0.013	0.6060 ± 0.015	0.6480 ± 0.014	0.6180 ± 0.012
	ResNet101+XLM-RoBERTa	0.7140 ± 0.012	0.6770 ± 0.013	0.7140 ± 0.012	0.6870 ± 0.011
	ResNet152+XLM-RoBERTa	0.6845 ± 0.013	0.6390 ± 0.014	0.6850 ± 0.013	0.6500 ± 0.012
	DenseNet121+XLM-RoBERTa	0.7340 ± 0.013	0.7210 ± 0.012	0.7320 ± 0.013	0.7260 ± 0.012
	DenseNet169+XLM-RoBERTa	0.7020 ± 0.013	0.6650 ± 0.015	0.7020 ± 0.012	0.6750 ± 0.013
	DenseNet201+XLM-RoBERTa	0.6670 ± 0.012	0.6200 ± 0.014	0.6670 ± 0.013	0.6330 ± 0.012
	MobileNetV2+XLM-RoBERTa	0.6780 ± 0.012	0.6410 ± 0.014	0.6780 ± 0.012	0.6510 ± 0.013
	MobileNetV3+XLM-RoBERTa	0.6820 ± 0.011	0.6450 ± 0.013	0.6820 ± 0.013	0.6550 ± 0.012
	ResNet50+DistilBERT	0.6860 ± 0.014	0.6490 ± 0.013	0.6860 ± 0.014	0.6590 ± 0.013
	ResNet101+DistilBERT	0.6900 ± 0.013	0.6530 ± 0.014	0.6900 ± 0.013	0.6630 ± 0.012
	ResNet152+DistilBERT	0.6940 ± 0.012	0.6570 ± 0.013	0.6940 ± 0.014	0.6670 ± 0.011
	DenseNet121+DistilBERT	0.6980 ± 0.012	0.6610 ± 0.013	0.6980 ± 0.013	0.6710 ± 0.012
	DenseNet169+DistilBERT	0.6640 ± 0.012	0.6180 ± 0.014	0.6640 ± 0.014	0.6300 ± 0.012
	DenseNet201+DistilBERT	0.7360 ± 0.011	0.7230 ± 0.012	0.7340 ± 0.012	0.7280 ± 0.011

Table 20
Performance metrics for multimodal-based models with intermediate fusion in Bangla fake news multiclass classification (5-fold cross-validation, Mean ± Std).

Approach	Model	Accuracy	Precision	Recall	F1-Score
Intermediate fusion	ResNet50+mBERT	0.7615 ± 0.011	0.7667 ± 0.012	0.7588 ± 0.013	0.7627 ± 0.010
	ResNet101+mBERT	0.7927 ± 0.009	0.8292 ± 0.010	0.7728 ± 0.011	0.8000 ± 0.009
	ResNet152+mBERT	0.7635 ± 0.010	0.7646 ± 0.011	0.7630 ± 0.012	0.7638 ± 0.010
	DenseNet121+mBERT	0.7635 ± 0.008	0.7792 ± 0.009	0.7556 ± 0.010	0.7672 ± 0.008
	DenseNet169+mBERT	0.7635 ± 0.009	0.7708 ± 0.008	0.7598 ± 0.009	0.7653 ± 0.009
	DenseNet201+mBERT	0.7594 ± 0.012	0.7688 ± 0.011	0.7546 ± 0.010	0.7616 ± 0.011
	MobileNetV2+mBERT	0.7615 ± 0.010	0.7646 ± 0.009	0.7598 ± 0.011	0.7622 ± 0.010
	MobileNetV3+mBERT	0.7563 ± 0.009	0.7625 ± 0.008	0.7531 ± 0.009	0.7578 ± 0.009
	ResNet50+XLM-RoBERTa	0.7854 ± 0.008	0.8146 ± 0.010	0.7697 ± 0.009	0.7915 ± 0.009
	ResNet101+XLM-RoBERTa	0.7823 ± 0.009	0.8125 ± 0.008	0.7662 ± 0.009	0.7887 ± 0.008
	ResNet152+XLM-RoBERTa	0.7490 ± 0.011	0.7563 ± 0.010	0.7454 ± 0.012	0.7508 ± 0.011
	DenseNet121+XLM-RoBERTa	0.7573 ± 0.008	0.7771 ± 0.009	0.7475 ± 0.009	0.7620 ± 0.009
	DenseNet169+XLM-RoBERTa	0.7677 ± 0.010	0.7938 ± 0.009	0.7545 ± 0.010	0.7736 ± 0.010
	DenseNet201+XLM-RoBERTa	0.7771 ± 0.009	0.8021 ± 0.008	0.7639 ± 0.009	0.7825 ± 0.008
	MobileNetV2+XLM-RoBERTa	0.7542 ± 0.008	0.7708 ± 0.010	0.7460 ± 0.008	0.7582 ± 0.009
	MobileNetV3+XLM-RoBERTa	0.7469 ± 0.010	0.7500 ± 0.009	0.7453 ± 0.011	0.7477 ± 0.010
	ResNet50+DistilBERT	0.7825 ± 0.009	0.7299 ± 0.010	0.7428 ± 0.009	0.7704 ± 0.009
	ResNet101+DistilBERT	0.7283 ± 0.012	0.7162 ± 0.013	0.7332 ± 0.011	0.7229 ± 0.012
	ResNet152+DistilBERT	0.7843 ± 0.008	0.7746 ± 0.009	0.7606 ± 0.010	0.7796 ± 0.008
	DenseNet121+DistilBERT	0.7742 ± 0.009	0.7249 ± 0.010	0.7813 ± 0.011	0.7531 ± 0.010
	DenseNet169+DistilBERT	0.7745 ± 0.011	0.7816 ± 0.009	0.7354 ± 0.010	0.7188 ± 0.011
	DenseNet201+DistilBERT	0.7282 ± 0.010	0.7441 ± 0.012	0.7754 ± 0.009	0.7788 ± 0.009
	MobileNetV2+DistilBERT	0.7106 ± 0.012	0.7508 ± 0.011	0.7434 ± 0.010	0.7277 ± 0.010
	MobileNetV3+DistilBERT	0.7196 ± 0.011	0.7370 ± 0.010	0.7853 ± 0.011	0.7358 ± 0.010

(73.30%), followed by MobileNet V2 + XLM-RoBERTa with an accuracy of 73.40% and an F1 Score of 72.60%. The weakest model, DistilBERT-based approaches, showed lower performance, indicating limitations in generalization.

7. Qualitative analysis

7.1. Error analysis in multimodal fake news binary classification

In this section, we provide an in-depth error analysis for specific instances where our model's predictions do not align with the ground truth labels. This analysis focuses on the text and visual features that may have contributed to misclassification, specifically differentiating between 'fake' (1) and 'non-fake' (0) news (see Fig. 7).

7.1.1. Image 1: Misclassification due to detailed textual and visual cues

Textual Features: The model misclassified Image 1 as real, likely due to the presence of numerous factual details and authoritative sources, such as the involvement of the Bangladesh Bank and the Savings Department. This article includes detailed legal references and procedural explanations, which are typically associated with legitimate news. The specificity and precision of the language, including mentions of laws and specific governmental actions, may have led the model to categorize it as genuine, overlooking the misinformation context. These elements, which often signify credibility, could have misled the model by masking the underlying misinformation.

Visual Features: The image associated with Image 1 depicts a Bangladeshi savings certificate with a Bengali text stating "National Savings Certificate" and "Bangladesh", along with a small stack of money. This visual representation is highly formal and official-looking, which may have further reinforced the perception of authenticity. The professional appearance and use of legitimate financial document likely contributed to the model's misclassification. The model may have interpreted formal and official imagery as indicators of real news, thus overlooking the potential for misinformation.

Table 21
Performance metrics for multimodal-based models with intermediate fusion in Bangla fake news multiclass classification (5-fold cross-validation, Mean ± Std).

Approach	Model	Accuracy	Precision	Recall	F1-Score
Intermediate fusion	ResNet50+mBERT	0.6800 ± 0.010	0.6840 ± 0.011	0.6870 ± 0.010	0.6850 ± 0.011
	ResNet101+mBERT	0.6870 ± 0.009	0.6900 ± 0.010	0.6930 ± 0.009	0.6910 ± 0.010
	ResNet152+mBERT	0.6930 ± 0.010	0.6970 ± 0.010	0.7000 ± 0.009	0.6980 ± 0.010
	DenseNet121+mBERT	0.7000 ± 0.008	0.7030 ± 0.009	0.7060 ± 0.010	0.7040 ± 0.009
	DenseNet169+mBERT	0.6670 ± 0.010	0.6710 ± 0.011	0.6740 ± 0.011	0.6720 ± 0.010
	DenseNet201+mBERT	0.6730 ± 0.010	0.6770 ± 0.010	0.6800 ± 0.009	0.6780 ± 0.010
	MobileNetV2+mBERT	0.7180 ± 0.009	0.6810 ± 0.010	0.7180 ± 0.009	0.6910 ± 0.010
	MobileNetV3+mBERT	0.6530 ± 0.010	0.6580 ± 0.009	0.6610 ± 0.009	0.6590 ± 0.010
	ResNet50+XLM-RoBERTa	0.6600 ± 0.009	0.6640 ± 0.010	0.6680 ± 0.009	0.6660 ± 0.009
	ResNet101+XLM-RoBERTa	0.6560 ± 0.010	0.6120 ± 0.010	0.6560 ± 0.010	0.6240 ± 0.010
	ResNet152+XLM-RoBERTa	0.6600 ± 0.010	0.6150 ± 0.011	0.6600 ± 0.010	0.6270 ± 0.010
	DenseNet121+XLM-RoBERTa	0.6480 ± 0.010	0.6060 ± 0.011	0.6480 ± 0.010	0.6180 ± 0.010
	DenseNet169+XLM-RoBERTa	0.7220 ± 0.008	0.6850 ± 0.010	0.7220 ± 0.008	0.6950 ± 0.009
	DenseNet201+XLM-RoBERTa	0.7260 ± 0.009	0.6890 ± 0.009	0.7260 ± 0.009	0.6990 ± 0.008
	MobileNetV2+XLM-RoBERTa	0.6700 ± 0.009	0.6500 ± 0.010	0.6600 ± 0.010	0.6550 ± 0.009
	MobileNetV3+XLM-RoBERTa	0.6800 ± 0.009	0.6600 ± 0.009	0.6700 ± 0.009	0.6650 ± 0.010
	ResNet50+DistilBERT	0.6900 ± 0.009	0.6700 ± 0.009	0.6800 ± 0.009	0.6750 ± 0.009
	ResNet101+DistilBERT	0.7000 ± 0.008	0.6800 ± 0.009	0.6900 ± 0.009	0.6850 ± 0.009
	ResNet152+DistilBERT	0.7100 ± 0.009	0.6900 ± 0.009	0.7000 ± 0.009	0.6950 ± 0.009
	DenseNet121+DistilBERT	0.6600 ± 0.010	0.6400 ± 0.010	0.6500 ± 0.010	0.6450 ± 0.010
	DenseNet169+DistilBERT	0.6500 ± 0.011	0.6300 ± 0.010	0.6400 ± 0.010	0.6350 ± 0.010

Table 22
Performance metrics for multimodal-based models with late fusion in Bangla fake news binary classification (5-fold cross-validation, Mean ± Std).

Approach	Model	Accuracy	Precision	Recall	F1-Score
Late fusion	ResNet50+mBERT	0.7792 ± 0.011	0.7667 ± 0.013	0.7863 ± 0.010	0.7764 ± 0.012
	ResNet101+mBERT	0.7813 ± 0.010	0.7708 ± 0.012	0.7872 ± 0.011	0.7789 ± 0.009
	ResNet152+mBERT	0.7260 ± 0.018	0.7229 ± 0.017	0.7275 ± 0.016	0.7252 ± 0.015
	DenseNet121+mBERT	0.7313 ± 0.015	0.7333 ± 0.014	0.7303 ± 0.013	0.7318 ± 0.012
	DenseNet169+mBERT	0.7813 ± 0.012	0.7667 ± 0.015	0.7897 ± 0.013	0.7780 ± 0.011
	DenseNet201+mBERT	0.7417 ± 0.017	0.7542 ± 0.016	0.7358 ± 0.015	0.7449 ± 0.014
	MobileNetV2+mBERT	0.7406 ± 0.014	0.7500 ± 0.015	0.7377 ± 0.014	0.7438 ± 0.013
	MobileNetV3+mBERT	0.7364 ± 0.016	0.7417 ± 0.015	0.7355 ± 0.014	0.7386 ± 0.015
	ResNet50+XLM-RoBERTa	0.7416 ± 0.014	0.7521 ± 0.013	0.7382 ± 0.012	0.7451 ± 0.013
	ResNet101+XLM-RoBERTa	0.7844 ± 0.010	0.7958 ± 0.011	0.7780 ± 0.010	0.7868 ± 0.010
	ResNet152+XLM-RoBERTa	0.7745 ± 0.012	0.7875 ± 0.011	0.7683 ± 0.012	0.7778 ± 0.011
	DenseNet121+XLM-RoBERTa	0.7396 ± 0.013	0.7542 ± 0.014	0.7328 ± 0.013	0.7433 ± 0.014
	DenseNet169+XLM-RoBERTa	0.7766 ± 0.011	0.7917 ± 0.010	0.7692 ± 0.011	0.7803 ± 0.012
	DenseNet201+XLM-RoBERTa	0.7635 ± 0.012	0.7813 ± 0.012	0.7545 ± 0.011	0.7677 ± 0.013
	MobileNetV2+XLM-RoBERTa	0.7563 ± 0.015	0.7667 ± 0.014	0.7510 ± 0.014	0.7588 ± 0.015
	MobileNetV3+XLM-RoBERTa	0.7448 ± 0.016	0.7438 ± 0.017	0.7453 ± 0.016	0.7445 ± 0.016
	ResNet50+DistilBERT	0.7198 ± 0.017	0.7496 ± 0.016	0.7127 ± 0.018	0.7827 ± 0.017
	ResNet101+DistilBERT	0.7307 ± 0.014	0.7629 ± 0.013	0.7349 ± 0.014	0.7516 ± 0.013
	ResNet152+DistilBERT	0.7537 ± 0.012	0.7248 ± 0.013	0.7875 ± 0.012	0.7719 ± 0.012
	DenseNet121+DistilBERT	0.7851 ± 0.009	0.7815 ± 0.010	0.7578 ± 0.009	0.7837 ± 0.010
	DenseNet169+DistilBERT	0.7171 ± 0.018	0.7257 ± 0.017	0.7136 ± 0.018	0.7360 ± 0.017
	DenseNet201+DistilBERT	0.7411 ± 0.015	0.7317 ± 0.016	0.7762 ± 0.015	0.7385 ± 0.014
	MobileNetV2+DistilBERT	0.7324 ± 0.016	0.7534 ± 0.017	0.7213 ± 0.016	0.7741 ± 0.016
	MobileNetV3+DistilBERT	0.7160 ± 0.018	0.7889 ± 0.016	0.7717 ± 0.017	0.7259 ± 0.017

7.1.2. Image 2: Influence of clickbait tactics and celebrity endorsement

Textual Features: The model classified Image 2 as real, likely owing to its health advice tone and lack of overtly sensational language typical of fake news. The description focuses on practical health tips related to clothing and cold weather, which can be easily mistaken for legitimate content. Detailed advice and medical suggestions may have been perceived as valuable and credible, thereby leading to misclassification. The headline, while seemingly unrelated to body content, may not have been flagged by the model as sufficiently misleading to categorize it as fake. The disconnect between the headline and body content is a common trait in clickbait but may not have been evident enough for the model to identify.

Visual Features: The image associated with Image 2, featuring a well-known Bangladeshi celebrity such as Jaya Ahsan, can significantly influence the model's perception. Celebrities are often used in clickbait to attract attention, and their presence can lend legitimacy to content. High-quality images of Jaya Ahsan in fashionable attire could make the article appear more credible and less like a clickbait. The professional appearance and association with a known public figure might have overshadowed the incongruity between the headline and article content, leading the model to misclassify it as real.

7.1.3. Image 3: Political misclassification amidst emotive language and visual context

Textual Features: The model mistakenly classified Image 3 as fake owing to its use of strong accusations and defensive rhetoric. Political content often includes emotive language and allegations, that can inadvertently trigger the model's fake news detection mechanisms. Terms such as ''lies'', ''fabricated'', and ''attempts'' attempts may have been flagged by the model as signals of misinformation. Furthermore, the detailed recounting of historical grievances and complex political issues may have been misinterpreted as conspiratorial or exaggerated,

Table 23
Performance metrics for multimodal-based models with late fusion in multiclass Bangla fake news detection (5-fold cross-validation, Mean ± Std).

Approach	Model	Accuracy	Precision	Recall	F1-Score
Late fusion	ResNet50+mBERT	0.7060 ± 0.014	0.6690 ± 0.016	0.7060 ± 0.014	0.6790 ± 0.015
	ResNet101+mBERT	0.7100 ± 0.013	0.6730 ± 0.015	0.7100 ± 0.013	0.6830 ± 0.014
	ResNet152+mBERT	0.6635 ± 0.017	0.6180 ± 0.018	0.6630 ± 0.017	0.6300 ± 0.016
	DenseNet121+mBERT	0.7200 ± 0.013	0.7000 ± 0.015	0.7100 ± 0.013	0.7050 ± 0.014
	DenseNet169+mBERT	0.7300 ± 0.012	0.7100 ± 0.013	0.7200 ± 0.013	0.7150 ± 0.013
	DenseNet201+mBERT	0.7330 ± 0.011	0.7200 ± 0.012	0.7300 ± 0.012	0.7250 ± 0.011
	MobileNetV2+mBERT	0.6670 ± 0.015	0.6220 ± 0.016	0.6670 ± 0.015	0.6340 ± 0.016
	MobileNetV3+mBERT	0.6705 ± 0.016	0.6250 ± 0.016	0.6710 ± 0.015	0.6370 ± 0.016
	ResNet50+XLM-RoBERTa	0.7220 ± 0.013	0.6850 ± 0.014	0.7220 ± 0.013	0.6950 ± 0.014
	ResNet101+XLM-RoBERTa	0.6600 ± 0.017	0.6400 ± 0.016	0.6500 ± 0.016	0.6450 ± 0.017
	ResNet152+XLM-RoBERTa	0.7100 ± 0.014	0.6900 ± 0.015	0.7000 ± 0.014	0.6950 ± 0.014
	DenseNet121+XLM-RoBERTa	0.6900 ± 0.015	0.6700 ± 0.016	0.6800 ± 0.015	0.6750 ± 0.015
	DenseNet169+XLM-RoBERTa	0.7300 ± 0.012	0.7320 ± 0.013	0.7340 ± 0.012	0.7330 ± 0.012
	DenseNet201+XLM-RoBERTa	0.6700 ± 0.016	0.6500 ± 0.017	0.6600 ± 0.016	0.6550 ± 0.016
	MobileNetV2+XLM-RoBERTa	0.7340 ± 0.012	0.7210 ± 0.013	0.7320 ± 0.012	0.7260 ± 0.012
	MobileNetV3+XLM-RoBERTa	0.6500 ± 0.018	0.6300 ± 0.017	0.6400 ± 0.017	0.6350 ± 0.017
	ResNet50+DistilBERT	0.6210 ± 0.019	0.5880 ± 0.018	0.6210 ± 0.018	0.5970 ± 0.019
	ResNet101+DistilBERT	0.6240 ± 0.018	0.5900 ± 0.017	0.6240 ± 0.017	0.5990 ± 0.018
	ResNet152+DistilBERT	0.6280 ± 0.017	0.5920 ± 0.017	0.6280 ± 0.016	0.6030 ± 0.017
	DenseNet121+DistilBERT	0.6320 ± 0.016	0.5950 ± 0.016	0.6320 ± 0.015	0.6060 ± 0.016
	DenseNet169+DistilBERT	0.6870 ± 0.014	0.6900 ± 0.015	0.6930 ± 0.014	0.6910 ± 0.014
	DenseNet201+DistilBERT	0.6775 ± 0.015	0.6320 ± 0.016	0.6780 ± 0.015	0.6430 ± 0.015
	MobileNetV2+DistilBERT	0.6810 ± 0.015	0.6350 ± 0.016	0.6810 ± 0.015	0.6470 ± 0.015
	MobileNetV3+DistilBERT	0.6800 ± 0.016	0.6600 ± 0.016	0.6700 ± 0.015	0.6650 ± 0.016

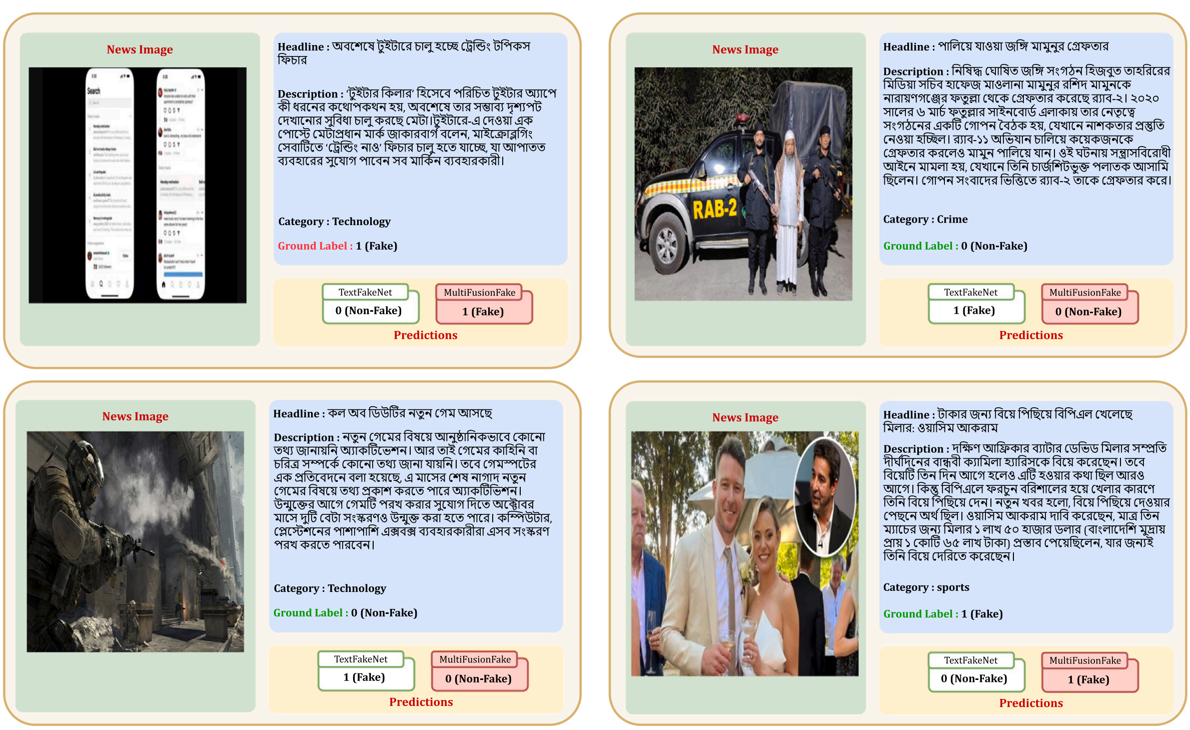

Fig. 6. Comparing the performance of the text-only model (TextFakeNet) and the multimodal early fusion model (MultiFusionFake), demonstrating the performance gain achieved by incorporating image features.

characteristics commonly associated with fake news. These combined factors likely led to the model's erroneous classification.

Visual Features: Visually, Image 3 includes political imagery, such as pictures of politicians, party symbols, or protest scenes, which can be associated with sensationalism. If the article included such imagery, it might have influenced the decision of the model. The use of official press release formats, logos of political parties, or images from political events might have been perceived as staged or manipulative, leading to false classifications. The visual elements could have been interpreted as indicative of propaganda, thereby triggering the fake news label. The image of Obaidul Quader, a prominent politician, could be seen as contributing to the model's misclassification due to the high stakes and emotive nature of political news imagery (see Fig. 8).

7.2. Error analysis in multimodal fake news multiclass classification

In this section, we provide an in-depth error analysis for specific instances where our model's predictions do not align with the ground truth labels. This analysis focuses on the text and visual features that may have contributed to misclassification, specifically differentiating between 'clickbait', 'misinformation', 'rumor', and 'non-fake' news.

Fig. 7. Visualization of error analysis in Bangla fake news detection for binary classification, highlighting misclassifications and key insights.

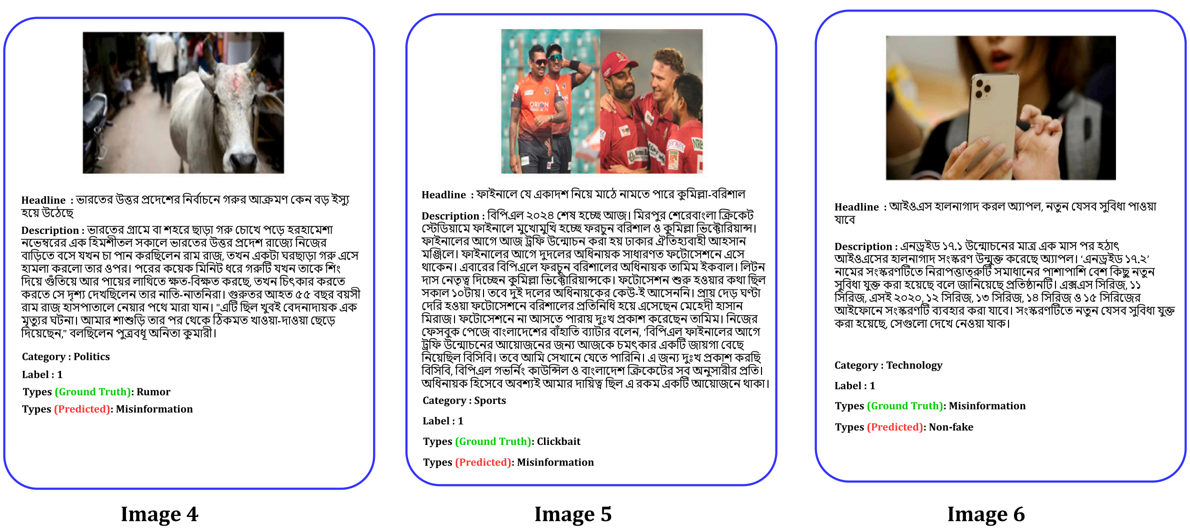

Fig. 8. Visualization of error analysis in Bangla fake news detection for multiclass classification, highlighting misclassifications across different news types.

7.2.1. *Image 4: Misclassifications in cattle breed identification*

Textual Features: "Boškarin", "Podolica", "Hallikar", "Nagori", "Umblachery", and "Gangatiri" are breed-specific names that represent particular cattle breeds. These tags suggest an attempt to categorize the bull based on perceived similarities to these breeds. In addition, the tags "Ox" and "Bull" are general classifications for bovine animals. The term "Ox" often refers to a castrated male, while "Bull" denotes an uncastrated adult male. These terms are less specific than the breed names and offer a broader classification. Another tag, "Krishna Valley", is a regional descriptor that indicates a geographic origin rather than a breed. Its inclusion in the tags might suggest a confusion between the location and the breed of the animal.

Visual Features: Visually, bulls are typically characterized by their heavily muscled bodies, especially in the shoulders and neck. The image likely displays these traits, and different breeds may exhibit variations in body build, such as stockier or leaner physiques, which might have influenced the breed tags. The shape, size, and direction of the horns are also crucial visual features for breed identification. The presence of horns in the image would have played a role in the classification, with long, curved horns potentially indicating certain breeds. The color and pattern of the bull's coat are significant visual cues as well. Different breeds possess distinct colorations, and the specific color of the bull in the image may have influenced the tagging process. Additionally, the facial features of the bull, such as the shape of the head, muzzle, and eyes, can vary across breeds, and these subtle differences could have been considered in the tagging process. Finally, the overall size and posture of the bull contribute to the impression formed, which likely influenced the identification and classification of the breed.

Relationship between Textual and Visual Features: The textual features (tags) are essentially interpretations of the visual features observed in the image. In this case, someone examined the bull in Image 4, analyzed its visual characteristics, and then applied the textual features, such as breed names, general terms, and regional descriptors, to describe what they saw. Errors arise when the interpretation, or the tag assigned, does not accurately reflect the visual reality, or when there is confusion regarding the meaning of certain textual features, such as the use of "Krishna Valley" as a breed name rather than a location. This disconnect between the visual characteristics and the assigned tags can lead to misclassification and errors in the tagging process.

7.2.2. *Image 5: Discrepancy between headline and content in BPL final report*

Textual Features: The textual features consist of the headline and the descriptive text. The headline mentions the BPL final, the teams involved (Comilla and Barishal), and the possibility of their playing eleven. The description elaborates on the events surrounding the final, including the trophy unveiling ceremony at Ahsan Manzil, the absence of the captains (Tamim Iqbal and Liton Das) from the photo session, and Tamim Iqbal's subsequent apology on Facebook. Specific textual elements include team names (Fortune Barishal, Comilla Victorians), player names (Tamim Iqbal, Liton Das, Mehdi Hasan Miraz), event names (BPL 2024, trophy unveiling), locations (Mirpur Sher-e-Bangla Cricket Stadium, Ahsan Manzil), organizations (BCB, BPL Governing Council), and time (10 am). The description also includes a direct quote from Tamim Iqbal expressing his regret.

Visual Features: Since we only have the headline and description, the visual features must be inferred. It is likely that the actual image accompanying this text would include team logos (Fortune Barishal and Comilla Victorians), the trophy being unveiled, and images of the players mentioned, particularly Tamim Iqbal, Liton Das, and Mehdi Hasan Miraz (perhaps at the photo session or in other contexts). The visual might also feature locations like Ahsan Manzil and/or the Sher-e-Bangla Cricket Stadium, and BPL branding, such as the BPL logo and other tournament-related visuals.

Relationship between Textual and Visual Features: The textual features provide context and meaning to the visual features. The headline and description tell the story of the BPL final, the trophy unveiling, and Tamim Iqbal's absence. The visual elements would likely illustrate these events. For instance, if the image shows the trophy, the text confirms it is the BPL 2024 trophy. If the image shows players, the text identifies them (e.g., Tamim Iqbal, Liton Das). The textual description of the photo session and Tamim's apology would be connected to any visual representation of those events.

7.2.3. Image 6: False misinformation label for factual iOS update report

Textual Features: The text describes a software update for Apple's iOS, specifically version 17.2. It provides detailed information about the update, including the specific version number and the devices it is compatible with, such as various iPhone models (XS, 11, SE 2020, 12, 13, 14, 15). This helps establish credibility and allows users to verify the information. The update is described in terms of security fixes and new features, which are typical elements in software update announcements. The language used is neutral, formal, and objective, with no sensationalism or emotional appeals, making the text sound factual and informative.

Visual Features: While the text itself does not include images, we can infer what visual elements might typically accompany such a report. An Apple logo would likely appear in the visuals to reinforce the source of the information. Images of the iPhone models mentioned would be included to show which devices are compatible with the update. Additionally, screenshots of the updated iOS interface might be included to illustrate the new features described in the text. A visual representation of the version number, "iOS 17.2", would likely be displayed to emphasize the specific update being discussed.

Relationship between Textual and Visual Features: The visual features would complement and support the textual information. The Apple logo would confirm the source of the update, and images of the compatible iPhone models would visually demonstrate which devices are eligible. Screenshots of the updated iOS interface would align with the textual descriptions of the new features, providing a clearer understanding of the changes. The visual reinforcement of the version number "iOS 17.2" would tie the textual details with the visual representation, further clarifying the specific update being discussed.

7.3. Computational efficiency

To assess the real-time applicability and training overhead of our proposed architecture, we evaluated inference latency, model size, floating-point operations (FLOPs), and training time for the MultiFusionFake (DenseNet-169+ mBERT) pipeline across the hardware configurations described in Section 6.1. All inference benchmarks in Table 24 were performed with a batch size of 1 and mixed-precision inference when supported.

While our DenseNet-169+ mBERT (MultiFusionFake) model delivers the highest accuracy (79.69%), its inference latency remains between 10–17 ms per sample on GPUs and exceeds 200 ms on CPUs. Furthermore, training the model for 30 epochs with a batch size of 8 takes between 8.1 and 10.75 h, depending on the hardware. These computational demands may pose challenges in scenarios with limited resources or strict time budgets.

7.4. Potential optimizations for deployment

To enable real-time deployment of our MultiFusionFake pipeline in resource-constrained environments, we explore potential optimization strategies aimed at reducing computational overhead without significantly compromising performance.

Knowledge Distillation: One promising direction is model compression via knowledge distillation, where a smaller "student" model is trained to replicate the behavior of the larger "teacher" model. This approach can reduce both inference latency and model size, making deployment feasible on edge devices or mobile platforms. For instance, distilled variants of BERT (e.g., DistilBERT) have shown substantial reductions in parameters and FLOPs while preserving most of the original model's accuracy.

Model Pruning: Another approach involves pruning redundant weights or entire channels/neurons in the network, particularly in the DenseNet backbone. Structured pruning techniques, which remove groups of parameters based on their contribution to output, can yield lightweight models that maintain acceptable accuracy with significantly fewer computations.

Quantization: Post-training quantization—converting 32-bit floating-point weights and activations to 8-bit integers—can further accelerate inference and reduce memory footprint, especially when deployed on hardware that supports INT8 acceleration.

Early Exit Mechanisms: Integrating early-exit classifiers into the network could allow faster inference for "easier" samples, enabling dynamic computation paths that reduce average latency while preserving performance for more complex inputs.

Lightweight Architectures: Replacing DenseNet-169 with a more compact CNN backbone (e.g., MobileNetV3, EfficientNet-B0) and substituting mBERT with lighter transformers (e.g., ALBERT, TinyBERT) could also drastically reduce model complexity. Preliminary experiments with these substitutes may serve as a foundation for future work.

Overall, incorporating these techniques holds promise for extending the applicability of our fusion model to real-time and embedded settings, which we plan to explore in follow-up studies.

8. Real-world applicability and deployment considerations

Deploying MultiFusionFake for real-time fake news detection requires balancing accuracy, efficiency, and scalability. Given its reliance on DenseNet 169 and mBERT, inference speed can be a concern, especially on resource-limited devices. To address this, techniques such as model quantization, knowledge distillation, and lightweight fusion strategies can enhance computational efficiency. For scalability, our model can be integrated into fact-checking systems, news verification pipelines, and social media moderation tools. Additionally, to adapt to evolving misinformation trends, incorporating online learning and active learning strategies will allow periodic updates with new data. Ethical considerations are also crucial, as false positives or negatives can impact public trust. A human-in-the-loop approach, where fact-checkers validate flagged content, can mitigate risks. By optimizing efficiency and adaptability, MultiFusionFake offers a promising step toward combating fake news in the Bangla language.

Table 24
Inference latency, parameter count, FLOPs, and 30-epoch training time for MultiFusionFake across different hardware platforms.

Hardware	Params (M)	FLOPs (G)	GPU latency (ms)	CPU latency (ms)	Training time (h)
RTX 4060 Ti (16 GB) + i7-9700F	138.4	45.2	11.8	235.4	8.10
RTX3050 (8 GB) + i5-9400F	138.4	45.2	16.9	312.7	10.75
Tesla P100 (12.7 GB) + Xeon (Kaggle)	138.4	45.2	15.5	280.2	9.60
Tesla T4 (15 GB) + Colab Xeon	138.4	45.2	13.4	260.1	9.05

9. Limitations

Although our research has made strides in the field of Multimodal Bangla Fake News Detection, there are certain limitations that warrant acknowledgment. First, we did not explore the use of romanized Bangla text in our detection models. Romanized Bangla refers to Bangla (Bengali) language text written using the Roman alphabet. Incorporating this variant could offer valuable insights and potentially improve model performance by expanding the range of textual representations. Furthermore, the interpretability of our models has not been thoroughly examined. Understanding how and why a model arrives at certain predictions is crucial for ensuring transparency and building user trust. Moreover, while our work focuses primarily on static multimodal content (text and images), we did not integrate methods for real-time content analysis or network-level mitigation strategies. Inspired by recent advances, we plan to incorporate techniques such as deep neural network-based harmful content detection and network immunization to contain the spread of misinformation, as proposed in STOPHC (Truică, Constantinescu, & Apostol, 2024). Additionally, we recognize the potential of integrating geolocation-content community detection and publisher–subscriber distribution models for real-time hazard or event tracking, which can be extended to improve fake news monitoring and dissemination control, as demonstrated in the distributed disaster reporting system (Apostol, Truică, & Paschke, 2024). These methods are currently absent in our framework but present promising directions for future enhancements. Furthermore, our research did not incorporate audio or video-based modalities, which could enrich the analysis by capturing additional linguistic and visual cues critical for fake news detection. Expanding to these modalities would allow us to handle a broader range of misinformation formats. Lastly, while MultiBanFakeDetect includes a diverse set of topics, its size (9600 text–image pairs) and scope may limit its generalizability across the full spectrum of Bangla dialects and evolving misinformation trends. Though we have strived for representative coverage, the dataset may not fully encapsulate regional linguistic variation or emerging narratives.

10. Future research directions

In our future research, we aim to significantly enhance the robustness, accuracy, and transparency of fake news detection in Bangla. One of the key directions we plan to pursue is integrating our detection model into proactive frameworks, such as network immunization systems. By implementing early detection, we can directly support strategies to limit the spread of misinformation, thereby reducing its impact. In addition to this, we will focus on adapting our model for real-time detection in dynamic environments, particularly on social media platforms where misinformation spreads rapidly. Optimizing the system for low-latency inference without compromising accuracy will be a critical challenge in this regard. Another important direction involves integrating our detection system with community-based approaches, which will enable crowdsourced fact-checking and collective verification. This collaborative method will strengthen the credibility of the system, as it allows the public to actively participate in the verification process. To further refine our detection capabilities, we will explore the use of tree-based models to improve classification performance. These structured models will not only enhance the decision-making process but also improve the system's interpretability and adaptability to a wide variety of content types. We will also focus on multimodal evidence retrieval by developing a re-ranking approach that utilizes both Large Language Models (LLMs) and Large Vision Language Models (LVLMs). By processing text and image-based evidence, this approach will allow for more accurate and contextually relevant fact verification. In parallel, we will work on expanding our synthetic data generation process by creating examples that closely resemble real-world fake content, including fake news articles and manipulated images. This will expose our models to a broader range of deceptive formats and improve their generalization and robustness against evolving misinformation tactics. Currently, our research focuses on categories such as misinformation, rumors, and clickbait. However, we plan to extend this taxonomy to include other forms of deception, such as satire, hoaxes, propaganda, fabricated content, manipulated media, and conspiracy theories. Each of these categories requires distinct detection strategies, and incorporating them into our system will make it more comprehensive and adaptable to the sociocultural context of Bangla. To ensure better linguistic coverage and adaptability to dynamic online content, we will expand the MultiBanFakeDetect dataset by incorporating region-specific dialects and continuously updating it with emerging misinformation trends. Transparency is a central priority in our design, and to achieve this, we will employ advanced Explainable AI (XAI) techniques. Methods such as GradCAM++, Faster ScoreCAM, and LayerCAM will be used to visualize and interpret the internal workings of our models, providing stakeholders with clear insights into how predictions are made and fostering trust in the system. Furthermore, we plan to explore various Vision Transformer (ViT) architectures, such as the Data-efficient Image Transformer (DeiT), Swin Transformer, and Swift Transformer. By leveraging self-attention mechanisms, these architectures will enhance the performance, efficiency, and interpretability of our multimodal fake news detection models. Finally, virality prediction will also be incorporated as a crucial component of our fake news detection models. By predicting the potential virality of misinformation, we can better understand how fake news spreads and its possible impact on the audience. This will allow for more proactive strategies to mitigate the harm caused by viral misinformation. These directions will guide our future research, helping us build a reliable, explainable, and scalable system for detecting fake news in the Bangla language across various formats and platforms.

11. Conclusion

In conclusion, our study tackles the crucial challenge of fake news detection in the Bangla language by introducing the MultiBanFakeDetect dataset, which integrates both textual and visual elements for improved detection accuracy. We explored multimodal approaches through early, late, and intermediate fusion techniques, demonstrating that combining text and image modalities significantly boosts performance. Our proposed model, MultiFusionFake, which employs an early fusion technique combining DenseNet 169 and mBERT, has the maximum accuracy of 79.69%. This not only outperforms text-only methods such as mBERT (73.13%), but also other multimodal fusion strategies. Late fusion strategies, such as ResNet 101 with XLM-RoBERTa, scored 78.44%, whereas intermediate fusion approaches, such as ResNet 101 with mBERT, scored 79.27%. These findings demonstrate MultiFusionFake's ability in combining visual and textual data to detect false news stories and counterfeit photos, outperforming both uni-modal and multi-modal fusion techniques. This MultiFusionFake outperforms both unimodal and other multimodal methods in detecting Bangla fake news. Through comprehensive preprocessing, model building, hyperparameter tuning, and evaluation, our findings offer valuable insights for enhancing fake news detection systems and future research in the Bangla context.

CRediT authorship contribution statement

Fatema Tuj Johora Faria: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Mukaffi Bin Moin:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Zayeed Hasan:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Md. Arafat Alam Khandaker:** Writing – original draft, Visualization, Validation, Methodology, Formal analysis. **Niful Islam:** Visualization, Validation, Resources. **Khan Md Hasib:** Writing – review & editing, Visualization, Supervision. **M.F. Mridha:** Writing – review & editing, Visualization, Supervision, Methodology.

Consent to publishing

All the participants have waived any rights to claim to the research work. All the authors have agreed to publish the research without any conflict.

Code availability

The codes and supplementary coding files for this study are open to everyone publicly at: GitHub[2] platform.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The dataset used in this research is publicly accessible through: https://data.mendeley.com/datasets/k5pbz9795f/1.

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